Label-Conditioned Robot Vision

A Baseline Study in Controlled Visual Generation for Robotics

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Abstract

Robotic systems rely heavily on vision modules to interpret and interact with the physical world. In many robotics pipelines, visual perception is not unconstrained; instead, it is guided by **explicit semantic priors** such as object categories, task labels, or scene identifiers. This project investigates **Label-Conditioned Robot Vision**, where symbolic class information is used to control visual generation, and studies its behavior as a **baseline perception module**.

Using Conditional Generative Adversarial Networks (cGANs), we model the mapping from **semantic labels → visual observations**, treating label conditioning as an abstraction of high-level robotic knowledge (e.g., object class hypotheses). The goal is not photorealistic synthesis, but rather to evaluate **controllability, training stability, failure modes, and scalability** of label-conditioned vision across increasing dataset complexity.

Experiments are conducted progressively on MNIST, Oxford-102 Flowers, and CUB-200 Birds to analyze how label-conditioned vision behaves under varying class cardinality, visual entropy, and background noise—conditions commonly encountered in robotic perception tasks.

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1. Motivation and Robotics Context

In Robotics and Autonomous Systems, perception modules often operate under **structured constraints**:

- Object categories are known or hypothesized (e.g., tool type, obstacle class)
- Tasks impose semantic expectations (e.g., “find a flower”, “detect a bird-like object”)
- Vision is one component within a larger autonomy stack

Rather than learning unconstrained image generation, this project focuses on **controlled visual generation**, where semantic labels explicitly guide the visual

output. This framing aligns with robotics use cases such as:

- Perception prototyping before full deployment
- Synthetic data generation for rare classes
- Studying failure modes in semantic vision systems
- Understanding how symbolic information influences visual representations

The project therefore reframes cGANs not as artistic generators, but as **robot vision primitives**.

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2. Problem Formulation

We model robot vision as a conditional generative process:

$$ z \sim \mathcal{N}(0, I), \quad y \in \{0,1\}^K $$

where:

- $ z $ is a latent noise vector representing stochastic visual variability
- $ y $ is a one-hot encoded semantic label (class prior)
- $ x $ is a generated visual observation (image)

The generator learns a mapping: $$ G: (z, y) \rightarrow x $$

The discriminator learns to distinguish real vs generated samples under the same conditioning: $$ D: (x, y) \rightarrow [0,1] $$

This formulation mirrors robotics perception settings where:

- $ y $ represents symbolic or semantic context
- $ x $ represents sensor observations
- The system must respect both realism and semantic alignment

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3. Mathematical Modeling

3.1 Conditional GAN Objective

The adversarial objective is given by:

$$ \min_G \max_D \; \mathbb{E}_{x,y \sim p_{\text{data}}}[\log D(x,y)] + \mathbb{E}_{z \sim p(z), y \sim p(y)}[\log(1 - D(G(z,y),y))] $$

This objective enforces two constraints critical to robot vision:

1. **Visual plausibility** (generated images resemble sensor data)
2. **Semantic consistency** (generated images match the conditioning label)

Binary Cross-Entropy (BCE) loss is used throughout, reflecting a standard and interpretable baseline for adversarial training.

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4. System Architecture (Robotics Perspective)

Generator (Vision Synthesis Module)

- Inputs: latent noise $ z $ + semantic label $ y $
- Outputs: synthetic visual observation $ x $
- Architecture scales with dataset complexity:
    - Fully connected layers for MNIST (28×28 grayscale)
    - Convolutional upsampling for Oxford-102 (64×64 RGB)
    - Deeper convolutional architecture for CUB-200 (96×96 RGB)

Discriminator (Vision Consistency Verifier)

- Inputs: image $ x $ + semantic label $ y $
- Output: realism score (real vs fake)
- Acts as a learned consistency check between visual structure and semantic label

This generator–discriminator pairing can be interpreted as:

> *A vision module producing hypotheses and a verifier enforcing semantic plausibility.*

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5. Execution Pipeline

The project follows a structured robotics-style experimental pipeline:

1. **Dataset Preparation**
    - Normalize and resize images to fixed sensor resolutions
    - Encode labels as one-hot semantic priors
2. **Model Initialization**

- Generator and Discriminator instantiated per dataset
- Architecture complexity matched to visual entropy

3. **Offline Training**
    - Adversarial training using Adam optimizer
    - Fixed noise + labels used for consistent monitoring
    - Training loops are intentionally marked “DO NOT RUN” unless retraining is desired
4. **Evaluation**
    - Qualitative inspection of generated samples
    - Quantitative metrics:
        - FID (distributional similarity)
        - Inception Score (class separability)
5. **Analysis**
    - Loss dynamics (G vs D imbalance)
    - Mode confusion and semantic leakage
    - Resolution and dataset bias effects

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6. Datasets as Perception Stress Tests

Each dataset is treated as a **robot vision stress test**, not just a benchmark:

- **MNIST**
  Sanity check for label-conditioned visual controllability.
- **Oxford-102 Flowers**
  Tests fine-grained category separation under class imbalance and color/texture variance.
- **CUB-200 Birds**
  High-entropy visual environment with background clutter, pose variation, and semantic overlap—analogous to real-world robotic perception challenges.

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7. Purpose of the Project

This project is intentionally positioned as a **baseline**. Its purpose is to:

- Validate label-conditioned vision as a controllable perception primitive
- Identify limitations before scaling to higher-capacity models
- Build intuition for semantic conditioning in robotic vision systems
- Serve as a precursor to:
    - embedding-based conditioning
    - diffusion models
    - world models
    - sim-to-real data generation pipelines

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8. Expected Deliverables

By the end of this project, the following deliverables are produced:

- A fully reproducible notebook implementing label-conditioned robot vision
- Trained generators and discriminators for all datasets
- Saved visual outputs demonstrating semantic control
- Quantitative evaluation metrics (FID, Inception Score)
- A structured analysis of:
    - training stability
    - failure modes
    - scalability limits

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9. Scope and Limitations

This work does **not** claim:

- Photorealistic image synthesis
- Deployment-ready perception systems
- Multi-modal sensor fusion

Instead, it deliberately focuses on **controlled, interpretable, and analyzable robot vision behavior** at low-to-moderate resolution.

Summary

Label-Conditioned Robot Vision reframes conditional generative modeling as a robotics perception problem. By emphasizing controllability, semantic grounding, and honest failure analysis, this project provides a solid foundation for more advanced robotic vision and world modeling systems.

Contributions

This project was developed as a collaborative effort, with clear role separation across modeling, experimentation, and analysis.
The following contributions are acknowledged based on implementation and documentation records:

Snavya Sai Munti Mudugu Badri Prasad

GitHub: https://github.com/snavya0309

- Reproduction of the original **CGAN baseline (Mirza & Osindero, 2014)** on the MNIST dataset
- Baseline experimentation and result analysis for MNIST
- Comparative discussion against the original CGAN formulation

Sushma Niresh

GitHub: https://github.com/Sushmaniresh

- Dataset handling and preprocessing pipelines
- Implementation of **label-conditioning mechanisms**
- Design and extension of **convolutional cGAN architectures** for Oxford-102 and CUB-200 datasets

Ayushman Mishra

GitHub: https://github.com/aymisxx

- Designed and implemented Generator and Discriminator architectures across datasets

- Executed end-to-end training, evaluation, and result generation
- Developed the mathematical formulation and analytical interpretation of the models
- Led system-level modeling and analysis from a **Robotics & Autonomous Systems** standpoint
- Managed repository structure, experimentation workflow, documentation, and reproducibility

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Setting up GPU.

In [1]:
```
import torch
print("Device:", torch.cuda.get_device_name(0))
```

```
Device: NVIDIA GeForce RTX 4060 Laptop GPU
```

MNIST Original Baseline

Important Modules

In [2]:
```
!pip install torch torchvision numpy matplotlib pillow tqdm scikit-learn --qui
```

Imports & Setup.

In [3]:
```
import os, random, json, numpy as np, matplotlib.pyplot as plt
from PIL import Image
from tqdm import tqdm
from sklearn.model_selection import train_test_split

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader

import torchvision
import torchvision.transforms as transforms
from torchvision.datasets import MNIST
from torchvision.utils import make_grid, save_image

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Using device:", DEVICE)

# reproducibility
torch.manual_seed(42)
random.seed(42)
np.random.seed(42)
```

```
Using device: cuda
```

MNIST Baseline: Robot Vision Sanity Check Configuration

This section defines the **baseline experimental configuration** for the MNIST dataset, which is used as a **sanity check for label-conditioned robot vision**.

MNIST serves as a low-dimensional, low-entropy visual environment where semantic labels (digits 0–9) have a direct and interpretable correspondence to visual structure. This setup allows us to validate whether a vision module can:

- Respect explicit semantic conditioning
- Produce visually consistent outputs under controlled settings
- Exhibit stable training dynamics before scaling to more complex environments

The parameters below define the **sensor resolution, semantic space, and learning constraints** for this baseline vision module.

Key design choices:

- **Latent Dimension (`z_dim`)** models stochastic visual variability analogous to sensor noise.
- **Label Dimension (`y_dim`)** represents discrete semantic priors available to a robot.
- **Image Dimension (`img_dim`)** reflects a flattened low-resolution camera input (28×28 grayscale).
- **Batch Size and Learning Rate** are selected to balance convergence stability and training efficiency for a baseline study.

This configuration intentionally prioritizes **interpretability and controllability** over visual realism, making it suitable for validating the core mechanics of label-conditioned robot vision before progressing to higher-resolution and higher-entropy datasets.

In [4]:
```
dataset     = "MNIST"
batch_size  = 100
z_dim       = 100
y_dim       = 10
img_dim     = 784          # 28×28x1 (Greyscale)
lr          = 0.0002
epochs      = 200
save_dir    = "./outputs/MNIST"
os.makedirs(save_dir, exist_ok=True)
```

Label-Conditioned Vision Architecture (MNIST Baseline)

This section defines the **Generator–Discriminator architecture** used to model **label-conditioned robot vision** in a low-resolution setting.

From a Robotics perspective, the Generator can be interpreted as a **vision synthesis module** that produces visual observations conditioned on explicit semantic priors, while the Discriminator acts as a **consistency verifier**, ensuring alignment between generated visuals and their corresponding semantic labels.

The MNIST baseline intentionally uses a **fully connected architecture** to maintain architectural simplicity and maximize interpretability. This design choice allows clear analysis of conditioning behavior without the added complexity of spatial feature learning.

Generator: Semantic-to-Visual Mapping Module

The Generator models the mapping:

$$ (z, y) \rightarrow x $$

where:

- $ z $ represents stochastic variability (analogous to sensor noise or environmental uncertainty),
- $ y $ represents explicit semantic information available to the robot (one-hot class label),
- $ x $ is the generated visual observation.

Key architectural features:

- **Separate processing paths** for latent noise and semantic labels, reflecting the separation between uncertainty and symbolic priors in robotic perception.
- **Feature-level fusion** via concatenation, enabling the model to jointly reason over stochastic and semantic inputs.
- **Batch Normalization and Dropout** to stabilize training and prevent overfitting in a low-capacity baseline model.
- **Sigmoid output activation**, constraining outputs to valid pixel intensity ranges.

Discriminator: Semantic Consistency Verifier

The Discriminator evaluates whether a given image–label pair is **semantically consistent and visually plausible**.

It receives:

- a flattened image vector (visual observation),
- the corresponding semantic label (conditioning signal),

and outputs a scalar probability indicating real vs generated data.

Key design considerations:

- Separate feature extraction for visual and semantic inputs, followed by late-stage fusion.
- Use of **LeakyReLU activations** to maintain gradient flow and stable adversarial dynamics.
- Dropout-based regularization to prevent discriminator overconfidence, which can destabilize adversarial training.

Architectural Scope and Limitations

This fully connected baseline:

- Does **not** preserve spatial locality, leading to blurred outputs,
- Is intentionally low-capacity to expose **conditioning behavior and failure modes** clearly,
- Serves as a **sanity check** before transitioning to convolutional architectures for higher-resolution robot vision tasks.

The instantiated Generator (`G`) and Discriminator (`D`) together form the **MNIST label-conditioned robot vision module**, which will be trained and evaluated in subsequent sections.

In [5]:
```
class Generator(nn.Module):
    def __init__(self, z_dim=100, y_dim=10, img_dim=784):
        super().__init__()
        self.y_hidden = nn.Sequential(
            nn.Linear(y_dim, 1000),
            nn.BatchNorm1d(1000), nn.ReLU(inplace=True), nn.Dropout(0.5)
        )
        self.z_hidden = nn.Sequential(
```

```
            nn.Linear(z_dim, 200),
            nn.BatchNorm1d(200), nn.ReLU(inplace=True), nn.Dropout(0.5)
        )
        self.combined = nn.Sequential(
            nn.Linear(1200, 1200), nn.BatchNorm1d(1200), nn.ReLU(inplace=True)
            nn.Linear(1200, img_dim), nn.Sigmoid()
        )

    def forward(self, z, y):
        z_proj = self.z_hidden(z)
        y_proj = self.y_hidden(y)
        return self.combined(torch.cat([z_proj, y_proj], dim=1))


class Discriminator(nn.Module):
    def __init__(self, img_dim=784, y_dim=10):
        super().__init__()
        self.x_hidden = nn.Sequential(
            nn.Linear(img_dim, 240),
            nn.BatchNorm1d(240), nn.LeakyReLU(0.2,inplace=True), nn.Dropout(0.
        )
        self.y_hidden = nn.Sequential(
            nn.Linear(y_dim, 50),
            nn.BatchNorm1d(50), nn.LeakyReLU(0.2,inplace=True), nn.Dropout(0.5
        )
        self.combined = nn.Sequential(
            nn.Linear(290, 240),
            nn.BatchNorm1d(240), nn.LeakyReLU(0.2,inplace=True), nn.Dropout(0.
            nn.Linear(240, 1), nn.Sigmoid()
        )

    def forward(self, x, y):
        x_proj = self.x_hidden(x)
        y_proj = self.y_hidden(y)
        return self.combined(torch.cat([x_proj, y_proj], dim=1))


# instantiate
G = Generator(z_dim, y_dim, img_dim).to(DEVICE)
D = Discriminator(img_dim, y_dim).to(DEVICE)
```

MNIST Data Ingestion and Semantic Encoding

This cell defines the **vision data pipeline** and **semantic label encoding** used for the MNIST baseline.

From a robotics perspective:

- The MNIST images are treated as **low-resolution camera observations**, flattened for compatibility with the fully connected

vision module.

- Class labels are converted to **explicit one-hot semantic representations**, serving as discrete priors available to the vision system.

This setup enables controlled evaluation of **label-conditioned visual generation**, where semantic information is injected directly into the vision model.

In [6]:
```
# MNIST Loader + one-hot util
transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Lambda(lambda x: x.view(-1)),  # flattening
])

mnist_train = MNIST(root="./data", train=True, download=True, transform=transf
train_loader = DataLoader(mnist_train, batch_size=batch_size, shuffle=True, dr

def make_one_hot(labels, num_classes):
    return torch.zeros(labels.size(0), num_classes, device=labels.device)\
                .scatter_(1, labels.unsqueeze(1), 1)
```

```
Downloading http://yann.lecun.com/exdb/mnist/train-images-idx3-ubyte.gz
Failed to download (trying next):
HTTP Error 404: Not Found

Downloading https://ossci-datasets.s3.amazonaws.com/mnist/train-images-idx3-uby
te.gz
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Downloading http://yann.lecun.com/exdb/mnist/train-labels-idx1-ubyte.gz
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HTTP Error 404: Not Found

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Downloading https://ossci-datasets.s3.amazonaws.com/mnist/t10k-labels-idx1-ubyte.gz
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Extracting ./data\MNIST\raw\t10k-labels-idx1-ubyte.gz to ./data\MNIST\raw
```

Offline Training Loop: Label-Conditioned Robot Vision (MNIST)

This cell implements the **offline learning procedure** for the MNIST label-conditioned robot vision module.

This process corresponds to training a vision subsystem that learns a mapping from **semantic priors (labels)** to **visual observations**, under adversarial supervision.

Key elements of the training setup:

- **Binary Cross-Entropy loss** is used to enforce realism and semantic consistency.
- **Adam optimization** with tuned momentum parameters stabilizes adversarial dynamics.
- The **Discriminator** is trained to verify consistency between images and their semantic labels.
- The **Generator** is trained to produce visual outputs that are both

realistic and label-aligned.

To enable consistent monitoring of vision behavior over time, a fixed set of latent vectors and semantic labels is used to periodically generate sample grids. This allows qualitative inspection of:

- semantic controllability,
- training progression,
- and early signs of mode confusion.

Model checkpoints, loss histories, and generated samples are saved to disk for post-training analysis.

In [7]:
```
criterion = nn.BCELoss()
G_opt     = optim.Adam(G.parameters(), lr=lr, betas=(0.5, 0.999))
D_opt     = optim.Adam(D.parameters(), lr=lr, betas=(0.5, 0.999))

# fixed noise & labels for monitoring
fixed_z      = torch.randn(y_dim * 10, z_dim, device=DEVICE)
fixed_y      = torch.arange(y_dim, device=DEVICE).repeat_interleave(10)
fixed_y_oh   = make_one_hot(fixed_y, y_dim)

D_losses, G_losses = [], []

for epoch in range(1, epochs + 1):
    for imgs, lbls in train_loader:
        imgs, lbls = imgs.to(DEVICE), lbls.to(DEVICE)
        b = imgs.size(0)

        real = torch.ones(b, 1, device=DEVICE)
        fake = torch.zeros(b, 1, device=DEVICE)
        y_oh = make_one_hot(lbls, y_dim)

        # Discriminator step
        D_opt.zero_grad()
        D_real = D(imgs, y_oh)
        z      = torch.randn(b, z_dim, device=DEVICE)
        D_fake = D(G(z, y_oh).detach(), y_oh)
        D_loss = criterion(D_real, real) + criterion(D_fake, fake)
        D_loss.backward()
        D_opt.step()

        # Generator step
        G_opt.zero_grad()
        z      = torch.randn(b, z_dim, device=DEVICE)
        G_loss = criterion(D(G(z, y_oh), y_oh), real)
        G_loss.backward()
        G_opt.step()

        D_losses.append(D_loss.item())
```

```
        G_losses.append(G_loss.item())

    # save sample grid every 40 epochs (and at epoch 1)
    if epoch == 1 or epoch % 40 == 0:
        with torch.no_grad():
            gen = G(fixed_z, fixed_y_oh).view(-1, 1, 28, 28)
        grid = make_grid(gen, nrow=10, normalize=True, value_range=(0,1))
        save_image(grid, os.path.join(save_dir, f"epoch_{epoch:03d}.png"))

    print(f"Epoch [{epoch}/{epochs}]  D_loss: {D_loss:.4f}  G_loss: {G_loss:.4

# save final models & losses
torch.save(G.state_dict(), os.path.join(save_dir, "G.pth"))
torch.save(D.state_dict(), os.path.join(save_dir, "D.pth"))
np.save(os.path.join(save_dir, "D_losses.npy"), D_losses)
np.save(os.path.join(save_dir, "G_losses.npy"), G_losses)
```

```
Epoch [1/200]  D_loss: 1.4144  G_loss: 0.6935
Epoch [2/200]  D_loss: 1.3664  G_loss: 0.7149
Epoch [3/200]  D_loss: 1.3281  G_loss: 0.7318
Epoch [4/200]  D_loss: 1.3009  G_loss: 0.7368
Epoch [5/200]  D_loss: 1.2910  G_loss: 0.8448
Epoch [6/200]  D_loss: 1.1776  G_loss: 0.8606
Epoch [7/200]  D_loss: 1.2824  G_loss: 0.7933
Epoch [8/200]  D_loss: 1.3408  G_loss: 0.9144
Epoch [9/200]  D_loss: 1.0777  G_loss: 0.9896
Epoch [10/200]  D_loss: 1.2975  G_loss: 0.9000
Epoch [11/200]  D_loss: 1.0854  G_loss: 1.0509
Epoch [12/200]  D_loss: 1.0771  G_loss: 1.0181
Epoch [13/200]  D_loss: 0.9864  G_loss: 1.1685
Epoch [14/200]  D_loss: 1.3070  G_loss: 1.1547
Epoch [15/200]  D_loss: 0.9676  G_loss: 1.1400
Epoch [16/200]  D_loss: 1.1862  G_loss: 1.1604
Epoch [17/200]  D_loss: 1.1194  G_loss: 1.2845
Epoch [18/200]  D_loss: 1.2513  G_loss: 1.0193
Epoch [19/200]  D_loss: 1.1434  G_loss: 0.9948
Epoch [20/200]  D_loss: 1.5491  G_loss: 0.8717
Epoch [21/200]  D_loss: 1.1559  G_loss: 1.5155
Epoch [22/200]  D_loss: 1.0959  G_loss: 1.2031
Epoch [23/200]  D_loss: 1.0019  G_loss: 1.2779
Epoch [24/200]  D_loss: 1.3801  G_loss: 1.1506
Epoch [25/200]  D_loss: 1.2901  G_loss: 1.2263
Epoch [26/200]  D_loss: 1.2503  G_loss: 0.9326
Epoch [27/200]  D_loss: 1.1811  G_loss: 1.0062
Epoch [28/200]  D_loss: 1.1099  G_loss: 1.3717
Epoch [29/200]  D_loss: 1.2156  G_loss: 0.9492
Epoch [30/200]  D_loss: 1.1949  G_loss: 1.1436
Epoch [31/200]  D_loss: 1.1072  G_loss: 0.8889
Epoch [32/200]  D_loss: 1.1591  G_loss: 1.0472
Epoch [33/200]  D_loss: 1.1089  G_loss: 1.0744
Epoch [34/200]  D_loss: 1.2778  G_loss: 0.7801
Epoch [35/200]  D_loss: 1.0713  G_loss: 1.4040
Epoch [36/200]  D_loss: 1.1702  G_loss: 1.0767
Epoch [37/200]  D_loss: 1.3098  G_loss: 1.0971
Epoch [38/200]  D_loss: 1.0479  G_loss: 0.9208
Epoch [39/200]  D_loss: 1.2225  G_loss: 0.9798
Epoch [40/200]  D_loss: 0.9761  G_loss: 0.8959
Epoch [41/200]  D_loss: 1.5671  G_loss: 0.9005
Epoch [42/200]  D_loss: 1.3245  G_loss: 0.8181
Epoch [43/200]  D_loss: 1.3196  G_loss: 1.5217
Epoch [44/200]  D_loss: 0.9929  G_loss: 0.9822
Epoch [45/200]  D_loss: 1.2282  G_loss: 0.9399
Epoch [46/200]  D_loss: 0.8232  G_loss: 0.9817
Epoch [47/200]  D_loss: 1.2030  G_loss: 0.8445
Epoch [48/200]  D_loss: 1.6299  G_loss: 1.0611
Epoch [49/200]  D_loss: 1.0845  G_loss: 1.2059
Epoch [50/200]  D_loss: 1.2616  G_loss: 0.9534
Epoch [51/200]  D_loss: 1.4123  G_loss: 1.2017
Epoch [52/200]  D_loss: 0.9634  G_loss: 1.0216
Epoch [53/200]  D_loss: 1.2511  G_loss: 1.2118
Epoch [54/200]  D_loss: 1.1807  G_loss: 0.9474
```

```
Epoch [55/200]  D_loss: 1.0538  G_loss: 1.1013
Epoch [56/200]  D_loss: 1.2956  G_loss: 1.1522
Epoch [57/200]  D_loss: 1.2961  G_loss: 0.8548
Epoch [58/200]  D_loss: 1.2793  G_loss: 0.8728
Epoch [59/200]  D_loss: 0.9893  G_loss: 1.1175
Epoch [60/200]  D_loss: 1.5013  G_loss: 0.8668
Epoch [61/200]  D_loss: 1.4004  G_loss: 0.9651
Epoch [62/200]  D_loss: 1.1095  G_loss: 1.0327
Epoch [63/200]  D_loss: 1.3940  G_loss: 0.9282
Epoch [64/200]  D_loss: 1.1280  G_loss: 0.9302
Epoch [65/200]  D_loss: 1.5412  G_loss: 0.9021
Epoch [66/200]  D_loss: 1.3912  G_loss: 0.9192
Epoch [67/200]  D_loss: 1.2449  G_loss: 0.8314
Epoch [68/200]  D_loss: 1.2294  G_loss: 0.9409
Epoch [69/200]  D_loss: 1.3394  G_loss: 0.9509
Epoch [70/200]  D_loss: 1.3897  G_loss: 0.6704
Epoch [71/200]  D_loss: 1.4573  G_loss: 0.8229
Epoch [72/200]  D_loss: 1.3122  G_loss: 0.9531
Epoch [73/200]  D_loss: 1.1872  G_loss: 1.1779
Epoch [74/200]  D_loss: 1.2206  G_loss: 0.8332
Epoch [75/200]  D_loss: 1.1079  G_loss: 0.9492
Epoch [76/200]  D_loss: 1.2858  G_loss: 0.7975
Epoch [77/200]  D_loss: 1.3864  G_loss: 1.0450
Epoch [78/200]  D_loss: 1.4566  G_loss: 0.9321
Epoch [79/200]  D_loss: 1.3718  G_loss: 0.9899
Epoch [80/200]  D_loss: 1.1303  G_loss: 0.7924
Epoch [81/200]  D_loss: 1.2440  G_loss: 0.9868
Epoch [82/200]  D_loss: 1.2238  G_loss: 0.9885
Epoch [83/200]  D_loss: 1.1831  G_loss: 0.8747
Epoch [84/200]  D_loss: 1.3138  G_loss: 0.9383
Epoch [85/200]  D_loss: 1.2335  G_loss: 0.8497
Epoch [86/200]  D_loss: 1.3524  G_loss: 0.7956
Epoch [87/200]  D_loss: 1.0996  G_loss: 0.7490
Epoch [88/200]  D_loss: 1.2629  G_loss: 0.9597
Epoch [89/200]  D_loss: 1.1249  G_loss: 1.0183
Epoch [90/200]  D_loss: 1.5411  G_loss: 0.9701
Epoch [91/200]  D_loss: 1.3707  G_loss: 1.0701
Epoch [92/200]  D_loss: 1.2530  G_loss: 0.9144
Epoch [93/200]  D_loss: 1.1786  G_loss: 0.9505
Epoch [94/200]  D_loss: 1.2143  G_loss: 0.9440
Epoch [95/200]  D_loss: 1.1082  G_loss: 0.8929
Epoch [96/200]  D_loss: 1.2678  G_loss: 1.0777
Epoch [97/200]  D_loss: 1.2784  G_loss: 0.9570
Epoch [98/200]  D_loss: 1.4285  G_loss: 0.9357
Epoch [99/200]  D_loss: 1.2489  G_loss: 1.0852
Epoch [100/200]  D_loss: 1.1485  G_loss: 0.9436
Epoch [101/200]  D_loss: 1.2141  G_loss: 0.7874
Epoch [102/200]  D_loss: 1.1477  G_loss: 0.8065
Epoch [103/200]  D_loss: 1.3501  G_loss: 0.8625
Epoch [104/200]  D_loss: 1.2033  G_loss: 0.9044
Epoch [105/200]  D_loss: 1.1068  G_loss: 0.7810
Epoch [106/200]  D_loss: 1.2266  G_loss: 0.7146
Epoch [107/200]  D_loss: 1.2735  G_loss: 0.8831
Epoch [108/200]  D_loss: 1.2269  G_loss: 0.7100
```

```
Epoch [109/200]  D_loss: 1.1254  G_loss: 0.8497
Epoch [110/200]  D_loss: 1.3542  G_loss: 0.7800
Epoch [111/200]  D_loss: 1.1914  G_loss: 0.7746
Epoch [112/200]  D_loss: 1.2999  G_loss: 0.9304
Epoch [113/200]  D_loss: 1.3075  G_loss: 0.9529
Epoch [114/200]  D_loss: 1.1643  G_loss: 0.9182
Epoch [115/200]  D_loss: 1.3854  G_loss: 0.9918
Epoch [116/200]  D_loss: 1.1752  G_loss: 0.9047
Epoch [117/200]  D_loss: 1.4511  G_loss: 0.8998
Epoch [118/200]  D_loss: 1.3398  G_loss: 0.8611
Epoch [119/200]  D_loss: 1.3682  G_loss: 0.7180
Epoch [120/200]  D_loss: 1.4491  G_loss: 0.9368
Epoch [121/200]  D_loss: 1.4686  G_loss: 0.8479
Epoch [122/200]  D_loss: 1.2224  G_loss: 0.8889
Epoch [123/200]  D_loss: 1.3020  G_loss: 0.8879
Epoch [124/200]  D_loss: 1.2664  G_loss: 0.9598
Epoch [125/200]  D_loss: 1.2755  G_loss: 0.8017
Epoch [126/200]  D_loss: 1.2210  G_loss: 0.7952
Epoch [127/200]  D_loss: 1.2799  G_loss: 0.9323
Epoch [128/200]  D_loss: 1.0722  G_loss: 0.9347
Epoch [129/200]  D_loss: 1.2713  G_loss: 0.9337
Epoch [130/200]  D_loss: 1.2819  G_loss: 1.0378
Epoch [131/200]  D_loss: 1.4235  G_loss: 0.9772
Epoch [132/200]  D_loss: 1.2895  G_loss: 0.9571
Epoch [133/200]  D_loss: 1.2523  G_loss: 1.0454
Epoch [134/200]  D_loss: 1.2912  G_loss: 1.0181
Epoch [135/200]  D_loss: 1.4977  G_loss: 0.8283
Epoch [136/200]  D_loss: 1.5162  G_loss: 0.7877
Epoch [137/200]  D_loss: 1.4755  G_loss: 0.9285
Epoch [138/200]  D_loss: 1.0497  G_loss: 1.0387
Epoch [139/200]  D_loss: 1.2534  G_loss: 0.8780
Epoch [140/200]  D_loss: 1.5414  G_loss: 0.7566
Epoch [141/200]  D_loss: 1.1999  G_loss: 1.0075
Epoch [142/200]  D_loss: 1.3108  G_loss: 1.0828
Epoch [143/200]  D_loss: 1.2690  G_loss: 1.1615
Epoch [144/200]  D_loss: 1.3118  G_loss: 0.8170
Epoch [145/200]  D_loss: 1.2440  G_loss: 0.9040
Epoch [146/200]  D_loss: 1.2398  G_loss: 0.8318
Epoch [147/200]  D_loss: 1.2091  G_loss: 1.0066
Epoch [148/200]  D_loss: 1.3420  G_loss: 0.6744
Epoch [149/200]  D_loss: 1.4340  G_loss: 0.7290
Epoch [150/200]  D_loss: 1.2431  G_loss: 0.8526
Epoch [151/200]  D_loss: 1.4629  G_loss: 0.7120
Epoch [152/200]  D_loss: 1.3650  G_loss: 0.8372
Epoch [153/200]  D_loss: 1.6282  G_loss: 0.9858
Epoch [154/200]  D_loss: 1.4760  G_loss: 0.9497
Epoch [155/200]  D_loss: 1.2663  G_loss: 0.8076
Epoch [156/200]  D_loss: 1.2221  G_loss: 0.9592
Epoch [157/200]  D_loss: 1.3934  G_loss: 1.0310
Epoch [158/200]  D_loss: 1.2293  G_loss: 1.0477
Epoch [159/200]  D_loss: 1.3876  G_loss: 0.7444
Epoch [160/200]  D_loss: 1.2474  G_loss: 0.8862
Epoch [161/200]  D_loss: 1.0998  G_loss: 0.7645
Epoch [162/200]  D_loss: 1.1901  G_loss: 0.7716
```

```
Epoch [163/200]  D_loss: 1.0432  G_loss: 1.0436
Epoch [164/200]  D_loss: 1.1225  G_loss: 1.0298
Epoch [165/200]  D_loss: 1.3527  G_loss: 0.8031
Epoch [166/200]  D_loss: 1.2942  G_loss: 0.9325
Epoch [167/200]  D_loss: 1.1937  G_loss: 0.8161
Epoch [168/200]  D_loss: 1.5010  G_loss: 0.9632
Epoch [169/200]  D_loss: 1.2359  G_loss: 0.7805
Epoch [170/200]  D_loss: 1.1995  G_loss: 0.7298
Epoch [171/200]  D_loss: 1.2357  G_loss: 0.8185
Epoch [172/200]  D_loss: 1.4615  G_loss: 0.9907
Epoch [173/200]  D_loss: 1.2486  G_loss: 0.8297
Epoch [174/200]  D_loss: 1.4573  G_loss: 0.8505
Epoch [175/200]  D_loss: 1.3381  G_loss: 1.0273
Epoch [176/200]  D_loss: 1.3674  G_loss: 0.9743
Epoch [177/200]  D_loss: 1.3666  G_loss: 0.9005
Epoch [178/200]  D_loss: 1.3093  G_loss: 0.9174
Epoch [179/200]  D_loss: 1.3616  G_loss: 0.8098
Epoch [180/200]  D_loss: 1.2366  G_loss: 0.8255
Epoch [181/200]  D_loss: 1.2874  G_loss: 0.8945
Epoch [182/200]  D_loss: 1.4361  G_loss: 1.0196
Epoch [183/200]  D_loss: 1.2141  G_loss: 0.9380
Epoch [184/200]  D_loss: 1.1404  G_loss: 0.6448
Epoch [185/200]  D_loss: 1.2804  G_loss: 0.8887
Epoch [186/200]  D_loss: 1.2165  G_loss: 0.9990
Epoch [187/200]  D_loss: 1.0977  G_loss: 1.0425
Epoch [188/200]  D_loss: 1.2482  G_loss: 0.8600
Epoch [189/200]  D_loss: 1.3046  G_loss: 0.8187
Epoch [190/200]  D_loss: 1.2290  G_loss: 0.6954
Epoch [191/200]  D_loss: 1.3487  G_loss: 0.8197
Epoch [192/200]  D_loss: 1.3263  G_loss: 0.8509
Epoch [193/200]  D_loss: 1.5068  G_loss: 0.8860
Epoch [194/200]  D_loss: 1.3519  G_loss: 0.8056
Epoch [195/200]  D_loss: 1.2681  G_loss: 0.9898
Epoch [196/200]  D_loss: 1.4588  G_loss: 0.8308
Epoch [197/200]  D_loss: 1.2631  G_loss: 0.7454
Epoch [198/200]  D_loss: 1.2321  G_loss: 1.0626
Epoch [199/200]  D_loss: 1.2364  G_loss: 0.8806
Epoch [200/200]  D_loss: 1.1366  G_loss: 1.0162
```

Qualitative Validation: Real vs Generated (Target Digits)

This cell performs a **qualitative sanity check** for the MNIST label-conditioned robot vision baseline by comparing real sensor-like observations to generated observations under the **same semantic constraint**.

For selected target classes (digits 3, 5, and 7), we:

- sample a small set of **real** images from the dataset,
- generate a matching number of **synthetic** images using the trained

Generator,

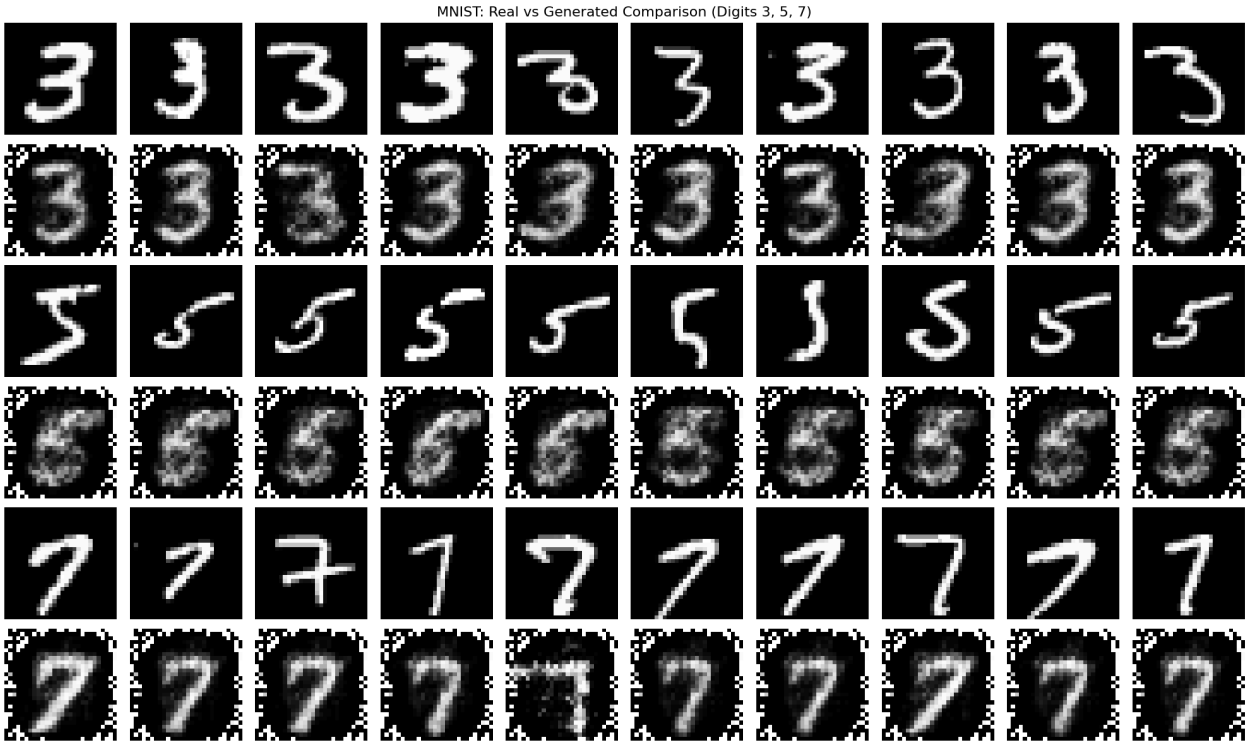
- visualize them side-by-side to evaluate **semantic controllability** and detect early failure modes such as **mode confusion** or class entanglement.

This is a robotics-style check: *does the vision module produce outputs consistent with the intended label prior?*

In [8]:
```
import torch
import matplotlib.pyplot as plt
from torchvision import datasets, transforms

# Settings
digits = [3, 5, 7]
num_samples = 10  # Number of real and fake samples per digit
save_dir = "./outputs/MNIST"

# Recreate Generator
G = Generator(z_dim, y_dim, img_dim).to(DEVICE)
G.load_state_dict(torch.load(f"{save_dir}/G.pth", map_location=DEVICE))
G.eval()

# One-hot encoding function
def make_one_hot(labels, num_classes):
    return torch.zeros(labels.size(0), num_classes, device=labels.device).scat

# Load Real MNIST Data
transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Lambda(lambda x: x.view(-1))  # Flatten to (784,)
])

mnist = datasets.MNIST(root="./data", train=True, download=True, transform=tra

# Collect real images for each digit
real_imgs_by_digit = {}
for d in digits:
    imgs = []
    for img, label in mnist:
        if label == d:
            imgs.append(img.view(28, 28))
        if len(imgs) >= num_samples:
            break
    real_imgs_by_digit[d] = torch.stack(imgs)

# Generate fake images for each digit
fake_imgs_by_digit = {}
for d in digits:
    z = torch.randn(num_samples, z_dim, device=DEVICE)
    labels = torch.full((num_samples,), d, dtype=torch.long, device=DEVICE)
    one_hot_labels = make_one_hot(labels, y_dim)
```

```
    with torch.no_grad():
        fake_imgs = G(z, one_hot_labels).cpu().view(-1, 28, 28)
    fake_imgs_by_digit[d] = fake_imgs

# Plot: for each digit -> 2 rows (Real, Generated)
rows = 2 * len(digits)
fig, axes = plt.subplots(rows, num_samples, figsize=(2 * num_samples, 2 * rows

for di, d in enumerate(digits):
    r_real = 2 * di
    r_fake = 2 * di + 1

    for i in range(num_samples):
        axes[r_real, i].imshow(real_imgs_by_digit[d][i], cmap="gray")
        axes[r_real, i].axis("off")
        if i == 0:
            axes[r_real, i].set_ylabel(f"Real {d}", fontsize=12)

    for i in range(num_samples):
        axes[r_fake, i].imshow(fake_imgs_by_digit[d][i], cmap="gray")
        axes[r_fake, i].axis("off")
        if i == 0:
            axes[r_fake, i].set_ylabel(f"Gen {d}", fontsize=12)

plt.suptitle("MNIST: Real vs Generated Comparison (Digits 3, 5, 7)", fontsize=
plt.tight_layout()
plt.show()
```

```
C:\Users\aymisxx\AppData\Local\Temp\ipykernel_10568\1585038710.py:12: FutureWar
ning: You are using `torch.load` with `weights_only=False` (the current default
value), which uses the default pickle module implicitly. It is possible to cons
truct malicious pickle data which will execute arbitrary code during unpickling
(See https://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models
for more details). In a future release, the default value for `weights_only` wi
ll be flipped to `True`. This limits the functions that could be executed durin
g unpickling. Arbitrary objects will no longer be allowed to be loaded via this
mode unless they are explicitly allowlisted by the user via `torch.serializatio
n.add_safe_globals`. We recommend you start setting `weights_only=True` for any
use case where you don't have full control of the loaded file. Please open an i
ssue on GitHub for any issues related to this experimental feature.
  G.load_state_dict(torch.load(f"{save_dir}/G.pth", map_location=DEVICE))
```



MNIST Outputs During Training: Semantic Controllability Check

The figure above shows **real vs generated MNIST samples** for digits **3, 5, and 7**, visualized while training is in progress.

Structure of the Visualization

- **Top row of each pair**: real MNIST samples corresponding to a fixed semantic label
- **Bottom row of each pair**: images generated by the vision module under the *same label condition*
- Each digit is evaluated independently to assess **semantic isolation and controllability**

Observations

- The generator consistently produces digit shapes aligned with the requested label (3, 5, or 7), indicating that **semantic conditioning is respected**.
- Generated samples are visibly blurrier than real images, which is expected given:
    - the fully connected architecture,

        - the low spatial resolution (28×28),
        - and the early-stage nature of training.
    - Despite reduced sharpness, **digit identity remains clear**, demonstrating that the model has learned class-level structure rather than memorizing individual samples.

Robotics Interpretation

In a robotics context, this result confirms that:

- symbolic priors (labels) can reliably control visual outputs,
- the perception module can generate **class-consistent observations** under uncertainty,
- and the core conditioning mechanism is functioning correctly.

This MNIST experiment establishes a **baseline level of semantic controllability**, validating the perception pipeline before transitioning to higher-resolution, higher-entropy environments such as Oxford-102 and CUB-200.

Transition to Real-World Vision

Oxford-102

Oxford-102 Configuration

This section defines the **system configuration** for extending label-conditioned robot vision from a low-entropy baseline (MNIST) to a **real-world, fine-grained visual dataset**.

The Oxford-102 Flowers dataset introduces challenges commonly encountered in robotic vision:

- high intra-class variability,
- inter-class visual similarity,
- class imbalance,
- and complex color and texture distributions.

This configuration simulates a **camera-based perception module** operating in a natural environment, where visual categories are more nuanced and less separable than synthetic or symbolic data.

Key configuration considerations:

- **Reduced batch size** to accommodate higher-resolution RGB imagery and increased model complexity.
- **Expanded semantic space (`y_dim = 102`)**, representing a richer set of object categories available to the vision system.
- **Fixed latent dimension** to maintain comparability with the MNIST baseline while isolating the effect of visual complexity.
- **Standardized image resolution (64×64 RGB)** to balance perceptual detail and computational feasibility.

This configuration sets the stage for evaluating how label-conditioned robot vision scales when exposed to **realistic visual diversity**, prior to further extension to even higher-entropy environments.

In [9]:
```
dataset_name = "Oxford102"

# Hyperparameters
batch_size  = 64            # Flowers are heavier than MNIST, so batch smaller
z_dim       = 100           # Latent vector dimension
y_dim       = 102           # 102 flower categories
img_size    = 64            # Resize all to 64x64
img_channels= 3             # RGB images
img_dim     = img_size * img_size * img_channels
lr          = 0.0002
epochs      = 200           # Same as MNIST

save_dir = f"./outputs/{dataset_name}"
os.makedirs(save_dir, exist_ok=True)

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")

print(f"Using device: {DEVICE}")
```

```
Using device: cuda
```

Oxford-102 Dataset Acquisition and Ground-Truth Setup

This cell prepares the **real-world visual dataset** used to evaluate label-conditioned robot vision beyond synthetic baselines.

This step corresponds to **sensor data provisioning and annotation loading**, where:

- raw visual observations (RGB images) are acquired,
- semantic ground-truth labels are retrieved separately,

- and both are organized into a structured dataset for downstream perception modeling.

Key aspects of this setup:

- The Oxford-102 Flowers dataset provides **high visual diversity** in color, texture, and shape, representative of natural environments encountered by robotic vision systems.
- Images and labels are downloaded independently, reflecting realistic scenarios where **sensor data and annotations are decoupled**.
- Labels stored in MATLAB format (`.mat`) are later parsed to construct explicit semantic priors for label-conditioned vision.

This step ensures a reproducible and well-defined dataset foundation before preprocessing, model training, and evaluation.

In [10]:
```python
import os
import urllib.request
import tarfile
import scipy.io

# Create target directory
oxford102_root = "./data/Oxford102"
os.makedirs(oxford102_root, exist_ok=True)

images_url = "https://www.robots.ox.ac.uk/~vgg/data/flowers/102/102flowers.tgz
labels_url = "https://www.robots.ox.ac.uk/~vgg/data/flowers/102/imagelabels.ma

# Target paths
images_tar_path = os.path.join(oxford102_root, "102flowers.tgz")
labels_mat_path = os.path.join(oxford102_root, "imagelabels.mat")

# Download images archive if not already downloaded
if not os.path.exists(images_tar_path):
    print("Downloading 102flowers.tgz...")
    urllib.request.urlretrieve(images_url, images_tar_path)
    print("Downloaded images archive!")

# Extract images if not already extracted
jpg_folder = os.path.join(oxford102_root, "jpg")
if not os.path.exists(jpg_folder):
    print("Extracting images...")
    with tarfile.open(images_tar_path, "r:gz") as tar:
        tar.extractall(path=oxford102_root)
    print("Extraction complete!")

# Download labels if not already downloaded
if not os.path.exists(labels_mat_path):
    print("Downloading imagelabels.mat...")
```

```
    urllib.request.urlretrieve(labels_url, labels_mat_path)
    print("Downloaded labels file!")

print("Oxford-102 dataset downloaded.")
```

```
Downloading 102flowers.tgz...
Downloaded images archive!
Extracting images...
Extraction complete!
Downloading imagelabels.mat...
Downloaded labels file!
Oxford-102 dataset downloaded.
```

Oxford-102 Dataset Interface and Vision Preprocessing

This section defines a **custom dataset interface** for integrating the Oxford-102 Flowers dataset into the label-conditioned robot vision pipeline.

This step corresponds to building a **vision sensor abstraction**, where raw RGB observations and semantic annotations are transformed into a standardized format consumable by the perception module.

Key design choices:

- **Image resizing** to a fixed resolution (64×64) simulates a constrained camera sensor input.
- **Normalization to [-1, 1]** ensures stable learning dynamics for generative vision models.
- **Explicit label alignment** converts dataset-provided annotations into zero-indexed semantic classes compatible with robot vision conditioning.
- A custom `Dataset` class enforces a clean separation between **data storage**, **preprocessing**, and **model consumption**, mirroring modular robotics software design.

The resulting DataLoader provides shuffled, batch-aligned image–label pairs, enabling scalable training and evaluation of label-conditioned vision under realistic visual diversity.

In [11]:
```
import os
import torch
from torch.utils.data import Dataset, DataLoader
from torchvision import transforms
from PIL import Image
import scipy.io
```

```python
# Define transformations
transform = transforms.Compose([
    transforms.Resize((img_size, img_size)),
    transforms.ToTensor(),
    transforms.Normalize([0.5, 0.5, 0.5], [0.5, 0.5, 0.5])
])

# Paths
oxford102_root = "./data/Oxford102"
image_dir = os.path.join(oxford102_root, "jpg")
labels_mat_path = os.path.join(oxford102_root, "imagelabels.mat")

# Custom Dataset Class
class Oxford102Dataset(Dataset):
    def __init__(self, image_dir, labels_mat_path, transform=None):
        self.image_dir = image_dir
        self.transform = transform

        # Load labels
        labels_data = scipy.io.loadmat(labels_mat_path)
        self.labels = labels_data["labels"][0]  # (8189,)

        # Load image file names
        self.image_files = sorted(os.listdir(image_dir))

        assert len(self.labels) == len(self.image_files), "Mismatch between nu

    def __len__(self):
        return len(self.image_files)

    def __getitem__(self, idx):
        img_path = os.path.join(self.image_dir, self.image_files[idx])
        image = Image.open(img_path).convert("RGB")
        label = self.labels[idx] - 1  # MATLAB indexing starts from 1 and not

        if self.transform:
            image = self.transform(image)

        return image, label

# Instantiate dataset and dataloader
oxford_dataset = Oxford102Dataset(
    image_dir=image_dir,
    labels_mat_path=labels_mat_path,
    transform=transform
)

oxford_loader = DataLoader(
    oxford_dataset,
    batch_size=batch_size,
    shuffle=True,
    drop_last=True,
    num_workers=0,
```

```
    pin_memory=True,
)

print(f"Oxford-102 dataset ready: {len(oxford_dataset)} images across {y_dim}
```

```
Oxford-102 dataset ready: 8189 images across 102 classes.
```

Oxford-102 Vision Module: Convolutional Label-Conditioned Architecture (64×64)

This cell defines the **64×64 RGB** label-conditioned generator and discriminator used for the Oxford-102 experiment.

In a robotics setting, moving from MNIST to Oxford-102 is a shift from a symbolic sanity check to a **camera-like perception regime**, where spatial structure (edges, textures, shapes) matters. Fully connected models blur spatial locality, so this stage uses a **convolutional (DCGAN-style) vision backbone** to preserve and refine spatial features during generation and verification.

Conditioning Mechanism: Spatial Semantic Injection

Oxford-102 uses **102 classes**, so the semantic prior y is high-dimensional. Instead of conditioning only at the input, this architecture injects semantics at **multiple spatial scales**:

- `expand_y(y, h, w)` tiles one-hot labels across the spatial grid
- The tiled label map is concatenated with feature maps at each conv stage

This mirrors robotics practice where **semantic context influences perception at different feature levels** (coarse shapes early, fine details later).

Generator64: Semantic-to-Image Synthesis (Camera Observation Model)

The generator produces an RGB observation:

$$ G(z, y) \rightarrow x \in \mathbb{R}^{3 \times 64 \times 64} $$

Design highlights:

- Latent z and label y are fused and projected into a compact spatial seed (4×4×512)

- A stack of transposed convolutions upsamples to 64×64
- Label maps are injected at every upsampling stage to enforce **semantic controllability**
- Final `tanh` output matches the normalized image range ([-1, 1])

Interpretation: a **label-conditioned camera simulator** producing class-aligned visual hypotheses.

Discriminator64: Conditional Realism + Consistency Verifier

The discriminator evaluates whether an image is both:

1. **visually plausible** (looks like a real flower image at this resolution), and
2. **consistent with the provided semantic prior** (label y)

$$ D(x, y) \rightarrow \text{logit} $$

Design highlights:

- The label map is concatenated at every downsampling stage
- Convolutional hierarchy extracts progressively higher-level visual structure
- Final output is a single logit (intended for **BCEWithLogitsLoss**)

Interpretation: a **semantic consistency verifier**—akin to checking whether a perception hypothesis matches a claimed class.

Why This Matters (Robotics Lens)

This architecture is the first “real” robot-vision version of the project:

- it preserves spatial structure,
- scales to 100+ semantic classes,
- and exposes robotics-relevant failure modes (class entanglement, background bias, limited detail at fixed resolution).

In [12]:
```
import torch.nn as nn
import torch

# Utility function to expand one-hot labels spatially
def expand_y(y, h, w):
```

```
    """Expand y: (B,C) -> (B,C,h,w) by spatial tiling."""
    return y.view(y.size(0), y.size(1), 1, 1).expand(-1, -1, h, w)

# Generator for 64x64 images
class Generator64(nn.Module):
    def __init__(self):
        super().__init__()
        self.fc = nn.Sequential(
            nn.Linear(z_dim + y_dim, 4 * 4 * 512),
            nn.BatchNorm1d(4 * 4 * 512),
            nn.ReLU(True)
        )
        self.conv_blocks = nn.ModuleList([
            nn.ConvTranspose2d(512 + y_dim, 256, 4, 2, 1, bias=False),  # 8x8
            nn.ConvTranspose2d(256 + y_dim, 128, 4, 2, 1, bias=False),  # 16x1
            nn.ConvTranspose2d(128 + y_dim, 64, 4, 2, 1, bias=False),   # 32x3
            nn.ConvTranspose2d(64 + y_dim, 3, 4, 2, 1, bias=False)      # 64x6
        ])
        self.bns = nn.ModuleList([
            nn.BatchNorm2d(256),
            nn.BatchNorm2d(128),
            nn.BatchNorm2d(64)
        ])

    def forward(self, z, y):  # z: (B, 100), y: (B, 102)
        x = torch.cat([z, y], dim=1)  # (B, 202)
        x = self.fc(x).view(-1, 512, 4, 4)  # (B, 512, 4, 4)

        for i, conv in enumerate(self.conv_blocks):
            h, w = x.shape[2], x.shape[3]
            y_exp = expand_y(y, h, w)
            x = torch.cat([x, y_exp], dim=1)
            x = conv(x)
            if i < 3:
                x = self.bns[i](x)
                x = nn.ReLU(True)(x)
            else:
                x = torch.tanh(x)
        return x

# Discriminator for 64x64 images
class Discriminator64(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv_blocks = nn.ModuleList([
            nn.Conv2d(3 + y_dim, 64, 4, 2, 1, bias=True),   # 32x32
            nn.Conv2d(64 + y_dim, 128, 4, 2, 1, bias=True), # 16x16
            nn.Conv2d(128 + y_dim, 256, 4, 2, 1, bias=True),# 8x8
            nn.Conv2d(256 + y_dim, 512, 4, 2, 1, bias=True) # 4x4
        ])
        self.fc = nn.Linear(4 * 4 * 512, 1)  # Final linear layer for logits

    def forward(self, x, y):  # x: (B, 3, 64, 64), y: (B, 102)
```

```
        for conv in self.conv_blocks:
            h, w = x.shape[2], x.shape[3]
            y_exp = expand_y(y, h, w)
            x = torch.cat([x, y_exp], dim=1)
            x = conv(x)
            x = nn.LeakyReLU(0.2, inplace=True)(x)

        x = x.view(x.size(0), -1)  # Flatten
        x = self.fc(x)             # Logits (use BCEWithLogitsLoss later)
        return x
```

Optimization Setup and Learning Objective (Oxford-102)

This cell initializes the **learning components** for the Oxford-102 label-conditioned robot vision module.

- The **Generator** and **Discriminator** together form a trainable perception subsystem.
- **Adam optimization** is used to ensure stable convergence under adversarial learning dynamics.
- **Binary Cross-Entropy with Logits** provides a numerically stable objective for evaluating visual realism and semantic consistency jointly.

This setup defines the **learning objective and control parameters** governing how the vision module adapts to complex, real-world visual data.

In [13]:
```
import torch.optim as optim
import torch.nn as nn

# Initialize models
G = Generator64().to(DEVICE)
D = Discriminator64().to(DEVICE)

# Optimizers
G_optimizer = optim.Adam(G.parameters(), lr=lr, betas=(0.5, 0.999))
D_optimizer = optim.Adam(D.parameters(), lr=lr, betas=(0.5, 0.999))

# Loss Function
criterion = nn.BCEWithLogitsLoss()

print("Optimizers and Loss function = READY")
```

```
Optimizers and Loss function = READY
```

Semantic Label Encoding Utility

This utility converts discrete class indices into **one-hot semantic vectors**, which serve as explicit conditioning signals for the robot vision module.

In a robotics context, this step formalizes symbolic category information into a machine-consumable representation, enabling controlled interaction between **semantic priors** and **visual generation**.

In [14]:
```
def make_one_hot(labels, num_classes):
    """
    Converts integer labels into one-hot encoded vectors.
    labels: (batch_size,) --> returns (batch_size, num_classes)
    """
    return torch.zeros(labels.size(0), num_classes, device=labels.device).scat
```

Visual Monitoring Utility for Perception Outputs

This helper function saves **gridded visual outputs** produced by the label-conditioned robot vision module at different training stages.

This serves as a **qualitative monitoring tool** to inspect:

- semantic controllability across labels,
- visual consistency within each class,
- and the progression of perceptual structure over training epochs.

Such visual checkpoints are essential for diagnosing perception failure modes that may not be immediately visible through scalar metrics alone.

In [15]:
```
import matplotlib.pyplot as plt

def save_image_grid(images, labels, epoch, save_dir, nrow=8):
    """
    Save a grid of generated images grouped by their labels.
    """
    images = images.detach().cpu()
    images = (images + 1) / 2  # Denormalize from [-1,1] to [0,1]

    batch_size = images.size(0)
    fig, axes = plt.subplots(nrows=(batch_size // nrow) + 1, ncols=nrow, figsi
    axes = axes.flatten()

    for img, ax in zip(images, axes):
        img = img.permute(1, 2, 0)  # (C,H,W) --> (H,W,C)
        ax.imshow(img)
        ax.axis('off')
```

```
    for ax in axes[batch_size:]:
        ax.axis('off')

    plt.tight_layout()
    os.makedirs(save_dir, exist_ok=True)
    plt.savefig(os.path.join(save_dir, f"epoch_{epoch:03d}.png"))
    plt.close()
```

Offline Training Loop: Oxford-102 Label-Conditioned Robot Vision

This section implements the **offline training procedure** for the Oxford-102 label-conditioned robot vision module.

This loop corresponds to training a **camera-based perception subsystem** under explicit semantic constraints. The system learns to associate **high-cardinality semantic labels (102 classes)** with realistic visual observations in a natural environment.

Key characteristics of this training process:

- The **Discriminator** evaluates both visual realism and semantic consistency between images and labels.
- The **Generator** learns to synthesize RGB observations that satisfy the provided semantic prior.
- Training proceeds in alternating optimization steps, reflecting adversarial learning dynamics.
- Fixed latent vectors and labels are used to generate **consistent visual checkpoints**, enabling longitudinal inspection of perceptual evolution.

Periodic visualization and saving of generated samples support:

- qualitative assessment of semantic controllability,
- detection of class entanglement and background bias,
- and analysis of convergence behavior beyond scalar loss values.

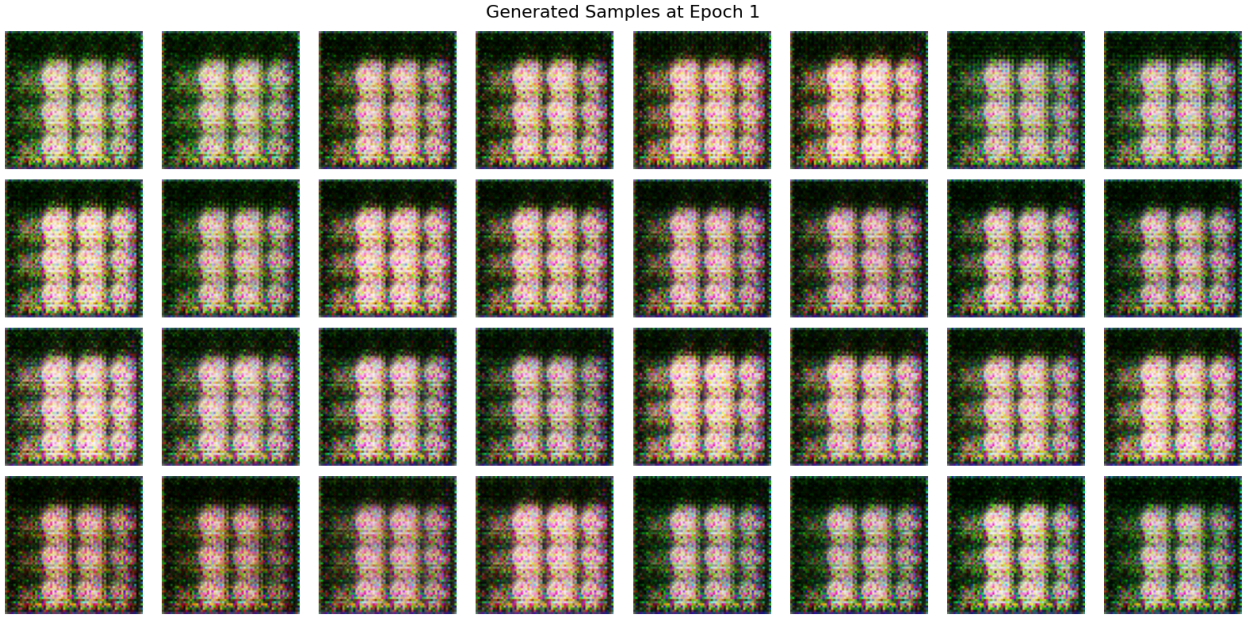
At the end of training, model parameters and loss histories are saved for reproducibility, post-hoc analysis, and comparison across datasets.

In [16]:
```
import numpy as np
from tqdm import tqdm
import matplotlib.pyplot as plt

# Fixed noise and labels for consistent evaluation
fixed_z = torch.randn(y_dim * 2, z_dim, device=DEVICE)
```

```
fixed_labels = torch.arange(y_dim, device=DEVICE).repeat_interleave(2)
fixed_labels_onehot = make_one_hot(fixed_labels, y_dim)

D_losses = []
G_losses = []

for epoch in range(1, epochs + 1):
    loop = tqdm(oxford_loader, leave=False, desc=f"Epoch {epoch}/{epochs}")
    for real_imgs, labels in loop:
        real_imgs, labels = real_imgs.to(DEVICE), labels.to(DEVICE)
        bsize = real_imgs.size(0)

        real_targets = torch.ones(bsize, 1, device=DEVICE)
        fake_targets = torch.zeros(bsize, 1, device=DEVICE)

        labels_onehot = make_one_hot(labels.long(), y_dim)

        # Train Discriminator
        D_optimizer.zero_grad()

        D_real = D(real_imgs, labels_onehot)
        D_loss_real = criterion(D_real, real_targets)

        z = torch.randn(bsize, z_dim, device=DEVICE)
        fake_imgs = G(z, labels_onehot)
        D_fake = D(fake_imgs.detach(), labels_onehot)
        D_loss_fake = criterion(D_fake, fake_targets)

        D_loss = D_loss_real + D_loss_fake
        D_loss.backward()
        D_optimizer.step()

        # Train Generator
        G_optimizer.zero_grad()

        z = torch.randn(bsize, z_dim, device=DEVICE)
        fake_imgs = G(z, labels_onehot)
        D_fake = D(fake_imgs, labels_onehot)
        G_loss = criterion(D_fake, real_targets)

        G_loss.backward()
        G_optimizer.step()

        D_losses.append(D_loss.item())
        G_losses.append(G_loss.item())


    print(f"Epoch [{epoch}/{epochs}]  D_loss: {D_loss.item():.4f}  G_loss: {G_

    # Save and Show Fake Samples every 40 epochs
    if epoch % 40 == 0 or epoch == 1:
        with torch.no_grad():
            fake_imgs_fixed = G(fixed_z, fixed_labels_onehot)
```

```
        # Save to file
        save_image_grid(fake_imgs_fixed, fixed_labels, epoch, save_dir)

        # Show in notebook
        fig, axes = plt.subplots(4, 8, figsize=(16, 8))
        axes = axes.flatten()
        fake_imgs_fixed = fake_imgs_fixed.detach().cpu()
        fake_imgs_fixed = (fake_imgs_fixed + 1) / 2  # denormalize

        for img, ax in zip(fake_imgs_fixed, axes):
            img = img.permute(1, 2, 0)
            ax.imshow(img)
            ax.axis('off')

        for ax in axes[len(fake_imgs_fixed):]:
            ax.axis('off')

        plt.suptitle(f"Generated Samples at Epoch {epoch}", fontsize=16)
        plt.tight_layout()
        plt.show()


np.save(os.path.join(save_dir, "D_losses.npy"), np.array(D_losses))
np.save(os.path.join(save_dir, "G_losses.npy"), np.array(G_losses))
torch.save(G.state_dict(), os.path.join(save_dir, "generator.pth"))
torch.save(D.state_dict(), os.path.join(save_dir, "discriminator.pth"))

print("Training done and all models and losses saved.")
```

```
Epoch [1/200]  D_loss: 1.6744  G_loss: 0.9149
```




```
Epoch [2/200]  D_loss: 1.4074  G_loss: 0.8049
```

```
Epoch [3/200]  D_loss: 1.3169  G_loss: 0.7801

Epoch [4/200]  D_loss: 1.3291  G_loss: 0.7430

Epoch [5/200]  D_loss: 1.3794  G_loss: 0.7952

Epoch [6/200]  D_loss: 1.3348  G_loss: 0.7731

Epoch [7/200]  D_loss: 1.3382  G_loss: 0.7471

Epoch [8/200]  D_loss: 1.3788  G_loss: 0.7320

Epoch [9/200]  D_loss: 1.3647  G_loss: 0.8409

Epoch [10/200]  D_loss: 1.3865  G_loss: 0.7325

Epoch [11/200]  D_loss: 1.3790  G_loss: 0.7281

Epoch [12/200]  D_loss: 1.3300  G_loss: 0.8415

Epoch [13/200]  D_loss: 1.3408  G_loss: 0.9897

Epoch [14/200]  D_loss: 1.2913  G_loss: 0.8562

Epoch [15/200]  D_loss: 1.3646  G_loss: 0.8150

Epoch [16/200]  D_loss: 1.3428  G_loss: 0.9257

Epoch [17/200]  D_loss: 1.3088  G_loss: 0.8065

Epoch [18/200]  D_loss: 1.3300  G_loss: 0.8016

Epoch [19/200]  D_loss: 1.2473  G_loss: 0.9310

Epoch [20/200]  D_loss: 1.1579  G_loss: 1.0961

Epoch [21/200]  D_loss: 1.1693  G_loss: 1.0576

Epoch [22/200]  D_loss: 1.2926  G_loss: 1.0303

Epoch [23/200]  D_loss: 1.2888  G_loss: 0.8284

Epoch [24/200]  D_loss: 1.3193  G_loss: 1.1688

Epoch [25/200]  D_loss: 1.2667  G_loss: 0.8878

Epoch [26/200]  D_loss: 1.3110  G_loss: 1.0324
```

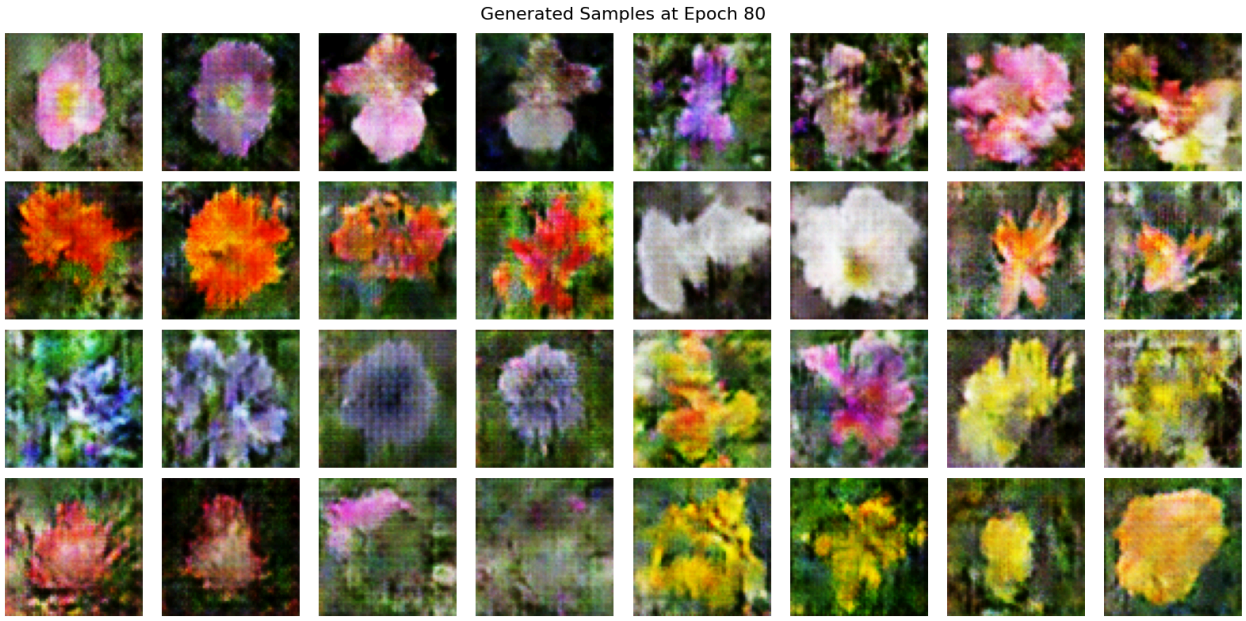
```
Epoch [27/200]  D_loss: 1.1081  G_loss: 0.9414

Epoch [28/200]  D_loss: 1.2058  G_loss: 1.1184

Epoch [29/200]  D_loss: 1.1227  G_loss: 1.0602

Epoch [30/200]  D_loss: 1.2226  G_loss: 1.1233

Epoch [31/200]  D_loss: 1.4097  G_loss: 1.2655

Epoch [32/200]  D_loss: 1.2151  G_loss: 1.0537

Epoch [33/200]  D_loss: 1.2252  G_loss: 1.3169

Epoch [34/200]  D_loss: 1.2238  G_loss: 1.3249

Epoch [35/200]  D_loss: 1.1166  G_loss: 1.1567

Epoch [36/200]  D_loss: 1.1080  G_loss: 1.1639

Epoch [37/200]  D_loss: 1.1043  G_loss: 1.0880

Epoch [38/200]  D_loss: 1.0155  G_loss: 1.4831

Epoch [39/200]  D_loss: 1.1812  G_loss: 1.2089

Epoch [40/200]  D_loss: 1.0785  G_loss: 1.2992
```

Generated Samples at Epoch 40

```
Epoch [41/200]  D_loss: 1.0904  G_loss: 1.4940
```

```
Epoch [42/200]  D_loss: 1.1243  G_loss: 1.5640
Epoch [43/200]  D_loss: 0.8749  G_loss: 1.3601
Epoch [44/200]  D_loss: 0.8583  G_loss: 1.5626
Epoch [45/200]  D_loss: 0.7941  G_loss: 1.5629
Epoch [46/200]  D_loss: 0.8972  G_loss: 1.4804
Epoch [47/200]  D_loss: 0.8341  G_loss: 1.9504
Epoch [48/200]  D_loss: 0.8503  G_loss: 2.2374
Epoch [49/200]  D_loss: 0.8837  G_loss: 1.3178
Epoch [50/200]  D_loss: 0.7455  G_loss: 1.7590
Epoch [51/200]  D_loss: 0.7866  G_loss: 1.9935
Epoch [52/200]  D_loss: 0.7864  G_loss: 2.5071
Epoch [53/200]  D_loss: 0.7767  G_loss: 1.7426
Epoch [54/200]  D_loss: 0.9216  G_loss: 2.2114
Epoch [55/200]  D_loss: 0.6264  G_loss: 2.3145
Epoch [56/200]  D_loss: 0.7361  G_loss: 2.3366
Epoch [57/200]  D_loss: 0.6872  G_loss: 2.1512
Epoch [58/200]  D_loss: 0.6409  G_loss: 2.1267
Epoch [59/200]  D_loss: 0.7075  G_loss: 2.0161
Epoch [60/200]  D_loss: 0.8227  G_loss: 2.2407
Epoch [61/200]  D_loss: 0.7855  G_loss: 1.9528
Epoch [62/200]  D_loss: 0.6985  G_loss: 2.2539
Epoch [63/200]  D_loss: 0.4995  G_loss: 2.8731
Epoch [64/200]  D_loss: 0.5789  G_loss: 3.0997
Epoch [65/200]  D_loss: 0.6091  G_loss: 3.2037
```

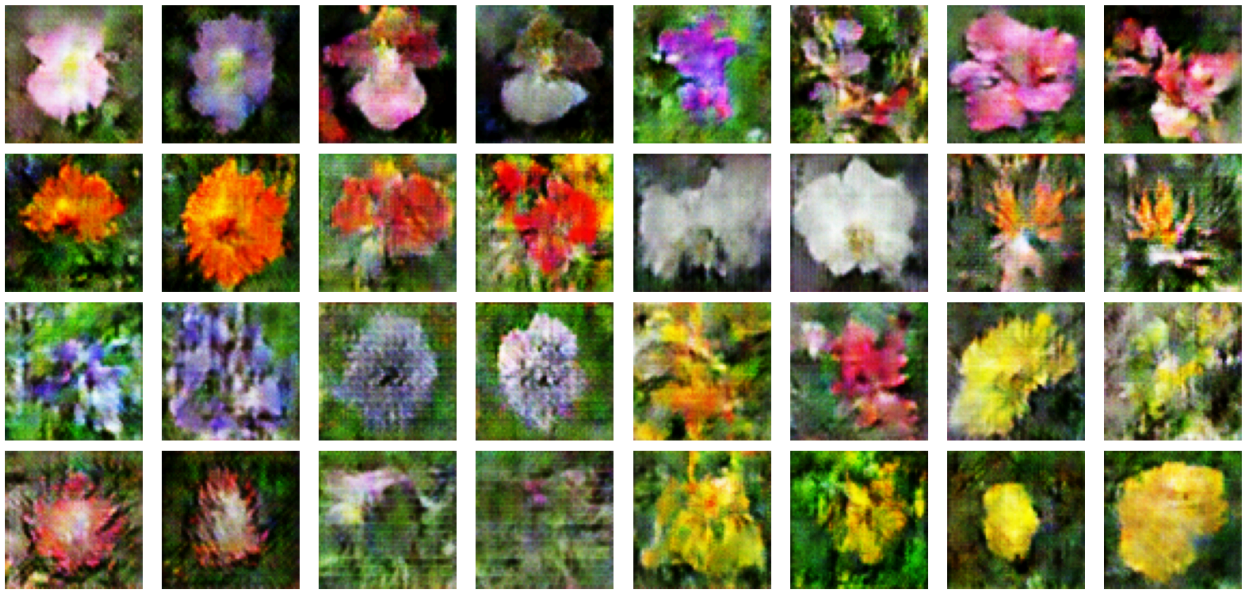
```
Epoch [66/200]  D_loss: 0.5444  G_loss: 2.5752

Epoch [67/200]  D_loss: 0.5394  G_loss: 2.4458

Epoch [68/200]  D_loss: 0.6112  G_loss: 2.2123

Epoch [69/200]  D_loss: 0.4577  G_loss: 2.8818

Epoch [70/200]  D_loss: 0.5608  G_loss: 3.4214

Epoch [71/200]  D_loss: 0.5067  G_loss: 3.4341

Epoch [72/200]  D_loss: 0.5791  G_loss: 2.5530

Epoch [73/200]  D_loss: 0.5108  G_loss: 2.5362

Epoch [74/200]  D_loss: 0.3986  G_loss: 2.8319

Epoch [75/200]  D_loss: 0.3792  G_loss: 3.1628

Epoch [76/200]  D_loss: 0.3792  G_loss: 3.5252

Epoch [77/200]  D_loss: 0.5449  G_loss: 2.6765

Epoch [78/200]  D_loss: 0.5998  G_loss: 2.9468

Epoch [79/200]  D_loss: 0.4639  G_loss: 2.9669

Epoch [80/200]  D_loss: 0.4382  G_loss: 2.9749
```



```
Epoch [81/200]  D_loss: 0.6957  G_loss: 2.7150

Epoch [82/200]  D_loss: 0.3303  G_loss: 3.3140

Epoch [83/200]  D_loss: 0.4186  G_loss: 3.6647

Epoch [84/200]  D_loss: 0.4667  G_loss: 4.2158

Epoch [85/200]  D_loss: 0.4554  G_loss: 3.4569

Epoch [86/200]  D_loss: 0.5253  G_loss: 3.0498

Epoch [87/200]  D_loss: 0.3779  G_loss: 3.8066

Epoch [88/200]  D_loss: 0.4254  G_loss: 3.7648

Epoch [89/200]  D_loss: 0.4845  G_loss: 3.8263

Epoch [90/200]  D_loss: 0.3527  G_loss: 4.6814

Epoch [91/200]  D_loss: 0.3976  G_loss: 3.5774

Epoch [92/200]  D_loss: 0.3333  G_loss: 4.2343

Epoch [93/200]  D_loss: 0.2962  G_loss: 3.2587

Epoch [94/200]  D_loss: 0.3457  G_loss: 3.1570

Epoch [95/200]  D_loss: 0.5370  G_loss: 3.2289

Epoch [96/200]  D_loss: 0.3040  G_loss: 3.7321

Epoch [97/200]  D_loss: 0.2133  G_loss: 4.1029

Epoch [98/200]  D_loss: 0.3432  G_loss: 3.7383

Epoch [99/200]  D_loss: 0.2399  G_loss: 4.2642

Epoch [100/200]  D_loss: 0.4860  G_loss: 4.3590

Epoch [101/200]  D_loss: 0.2690  G_loss: 4.3113

Epoch [102/200]  D_loss: 0.2599  G_loss: 4.6267

Epoch [103/200]  D_loss: 0.3427  G_loss: 4.2130

Epoch [104/200]  D_loss: 0.3264  G_loss: 4.4891
```

```
Epoch [105/200]  D_loss: 0.2722  G_loss: 4.7731

Epoch [106/200]  D_loss: 0.6062  G_loss: 5.4061

Epoch [107/200]  D_loss: 0.2910  G_loss: 3.1200

Epoch [108/200]  D_loss: 0.4831  G_loss: 4.5101

Epoch [109/200]  D_loss: 0.3867  G_loss: 3.7125

Epoch [110/200]  D_loss: 0.2998  G_loss: 5.6148

Epoch [111/200]  D_loss: 0.2364  G_loss: 4.7397

Epoch [112/200]  D_loss: 0.2548  G_loss: 5.6481

Epoch [113/200]  D_loss: 0.2604  G_loss: 4.9229

Epoch [114/200]  D_loss: 0.3027  G_loss: 5.2965

Epoch [115/200]  D_loss: 0.3240  G_loss: 5.8827

Epoch [116/200]  D_loss: 0.3270  G_loss: 5.6772

Epoch [117/200]  D_loss: 0.3421  G_loss: 4.4607

Epoch [118/200]  D_loss: 0.2699  G_loss: 4.9527

Epoch [119/200]  D_loss: 0.2461  G_loss: 5.0060

Epoch [120/200]  D_loss: 0.3215  G_loss: 4.9775
```

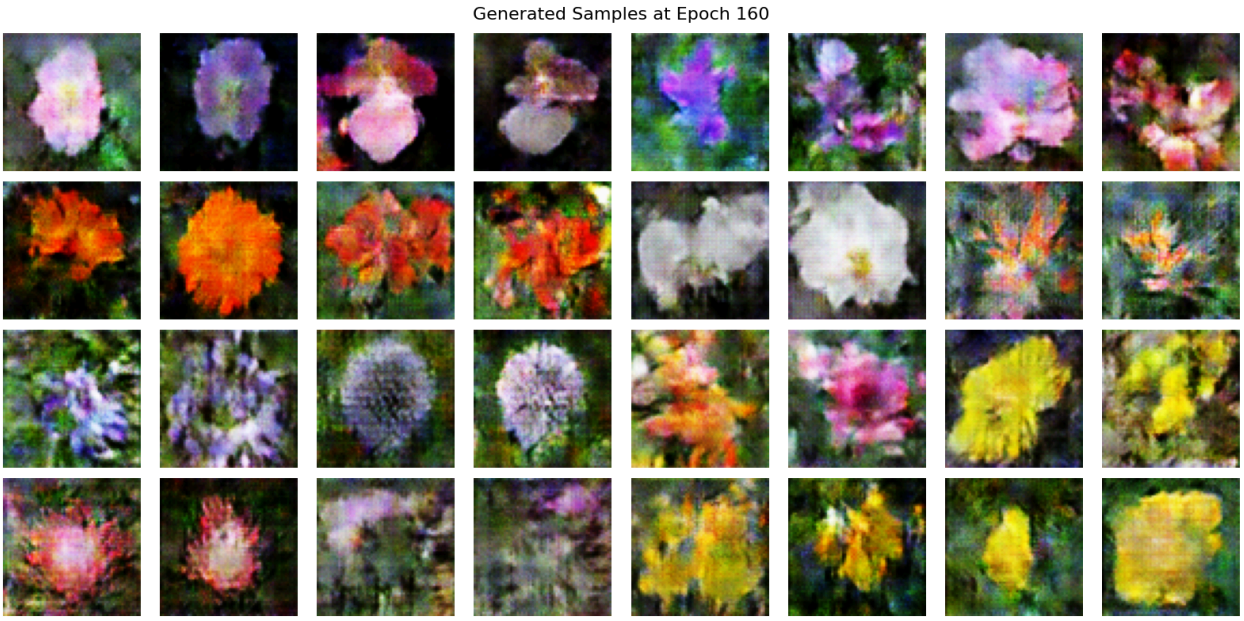
Generated Samples at Epoch 120



```
Epoch [121/200]  D_loss: 0.2650  G_loss: 5.0381

Epoch [122/200]  D_loss: 0.2707  G_loss: 5.4060

Epoch [123/200]  D_loss: 0.1408  G_loss: 5.8613

Epoch [124/200]  D_loss: 0.3450  G_loss: 5.0752

Epoch [125/200]  D_loss: 0.3134  G_loss: 5.4500

Epoch [126/200]  D_loss: 0.5105  G_loss: 4.6008

Epoch [127/200]  D_loss: 0.2348  G_loss: 6.8839

Epoch [128/200]  D_loss: 0.1869  G_loss: 4.8770

Epoch [129/200]  D_loss: 0.2755  G_loss: 5.8641

Epoch [130/200]  D_loss: 0.1944  G_loss: 4.7481

Epoch [131/200]  D_loss: 0.3512  G_loss: 6.2806

Epoch [132/200]  D_loss: 0.1354  G_loss: 5.4089

Epoch [133/200]  D_loss: 0.3126  G_loss: 5.5113

Epoch [134/200]  D_loss: 0.3319  G_loss: 6.3943

Epoch [135/200]  D_loss: 0.2796  G_loss: 4.5965
```

```
Epoch [136/200]  D_loss: 0.2838  G_loss: 4.8395

Epoch [137/200]  D_loss: 0.3922  G_loss: 4.9618

Epoch [138/200]  D_loss: 0.2456  G_loss: 5.3833

Epoch [139/200]  D_loss: 0.2311  G_loss: 5.9761

Epoch [140/200]  D_loss: 0.1916  G_loss: 6.0090

Epoch [141/200]  D_loss: 0.2491  G_loss: 5.3481

Epoch [142/200]  D_loss: 0.4080  G_loss: 6.0100

Epoch [143/200]  D_loss: 0.1364  G_loss: 4.7630

Epoch [144/200]  D_loss: 0.4429  G_loss: 5.8934

Epoch [145/200]  D_loss: 0.1348  G_loss: 5.7268

Epoch [146/200]  D_loss: 0.1308  G_loss: 6.0488

Epoch [147/200]  D_loss: 0.3158  G_loss: 5.8213

Epoch [148/200]  D_loss: 0.0995  G_loss: 6.4835

Epoch [149/200]  D_loss: 0.3052  G_loss: 4.4298

Epoch [150/200]  D_loss: 0.2615  G_loss: 5.7412

Epoch [151/200]  D_loss: 0.1201  G_loss: 6.2303

Epoch [152/200]  D_loss: 0.1525  G_loss: 6.5351

Epoch [153/200]  D_loss: 0.1342  G_loss: 5.3301

Epoch [154/200]  D_loss: 0.1423  G_loss: 6.5763

Epoch [155/200]  D_loss: 0.2325  G_loss: 5.5820

Epoch [156/200]  D_loss: 0.2314  G_loss: 6.2361

Epoch [157/200]  D_loss: 0.1712  G_loss: 5.9265

Epoch [158/200]  D_loss: 0.2423  G_loss: 6.0525

Epoch [159/200]  D_loss: 0.0926  G_loss: 6.2961
```

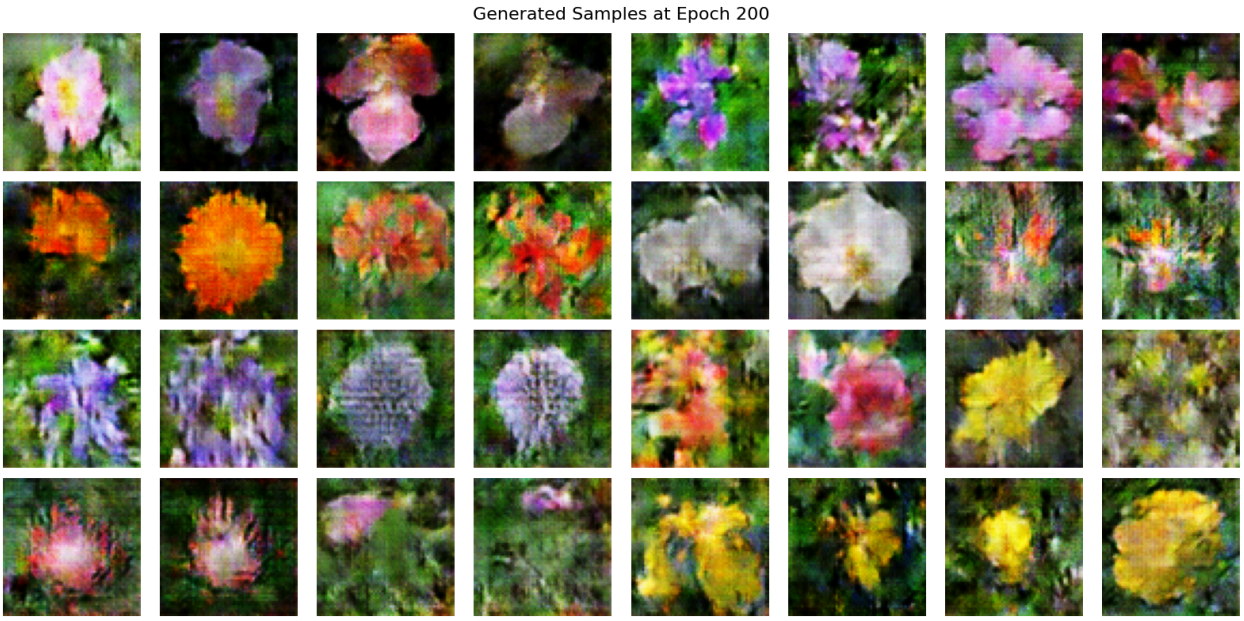
```
Epoch [160/200]  D_loss: 0.1030  G_loss: 6.2695
```




```
Epoch [161/200]  D_loss: 0.1588  G_loss: 8.1462

Epoch [162/200]  D_loss: 0.2745  G_loss: 6.5359

Epoch [163/200]  D_loss: 0.0738  G_loss: 6.3661

Epoch [164/200]  D_loss: 0.2742  G_loss: 6.7502

Epoch [165/200]  D_loss: 0.0843  G_loss: 6.3409

Epoch [166/200]  D_loss: 0.2413  G_loss: 6.5050

Epoch [167/200]  D_loss: 0.2480  G_loss: 6.7263

Epoch [168/200]  D_loss: 0.1028  G_loss: 5.0869

Epoch [169/200]  D_loss: 0.1092  G_loss: 6.4806

Epoch [170/200]  D_loss: 0.2625  G_loss: 7.5048

Epoch [171/200]  D_loss: 0.1801  G_loss: 6.1094

Epoch [172/200]  D_loss: 0.2767  G_loss: 8.1775

Epoch [173/200]  D_loss: 0.1612  G_loss: 5.4482

Epoch [174/200]  D_loss: 0.1369  G_loss: 7.3774
```

```
Epoch [175/200]  D_loss: 0.1724  G_loss: 7.1499

Epoch [176/200]  D_loss: 0.1948  G_loss: 6.6833

Epoch [177/200]  D_loss: 0.1792  G_loss: 6.8618

Epoch [178/200]  D_loss: 0.2338  G_loss: 6.8220

Epoch [179/200]  D_loss: 0.1374  G_loss: 5.7758

Epoch [180/200]  D_loss: 0.0929  G_loss: 7.2533

Epoch [181/200]  D_loss: 0.1295  G_loss: 5.8109

Epoch [182/200]  D_loss: 0.2190  G_loss: 6.9393

Epoch [183/200]  D_loss: 0.2358  G_loss: 7.4280

Epoch [184/200]  D_loss: 0.2012  G_loss: 7.4161

Epoch [185/200]  D_loss: 0.0975  G_loss: 6.8985

Epoch [186/200]  D_loss: 0.2160  G_loss: 7.3020

Epoch [187/200]  D_loss: 0.1101  G_loss: 7.7613

Epoch [188/200]  D_loss: 0.2072  G_loss: 7.4766

Epoch [189/200]  D_loss: 0.0486  G_loss: 6.8799

Epoch [190/200]  D_loss: 0.1815  G_loss: 8.6646

Epoch [191/200]  D_loss: 0.1232  G_loss: 8.3961

Epoch [192/200]  D_loss: 0.1612  G_loss: 8.6615

Epoch [193/200]  D_loss: 0.1224  G_loss: 7.1139

Epoch [194/200]  D_loss: 0.0780  G_loss: 8.0693

Epoch [195/200]  D_loss: 0.3180  G_loss: 7.8121

Epoch [196/200]  D_loss: 0.1729  G_loss: 5.5106

Epoch [197/200]  D_loss: 0.1106  G_loss: 7.6806

Epoch [198/200]  D_loss: 0.1523  G_loss: 7.6056
```

```
Epoch [199/200]  D_loss: 0.1078  G_loss: 7.0692

Epoch [200/200]  D_loss: 0.1733  G_loss: 7.2183
```




```
Training done and all models and losses saved.
```

Semantic Visualization: Class-Wise Vision Outputs (Oxford-102)

This section performs a **class-wise qualitative evaluation** of the trained label-conditioned robot vision module on the Oxford-102 dataset.

This visualization tests whether the vision system can:

- generate **distinct visual hypotheses** for each semantic class,
- maintain **semantic alignment** between labels and visual structure,
- and exhibit reasonable inter-class separation across a large category space.

One synthetic image is generated per class using fixed latent vectors and explicit semantic conditioning. The generated outputs are displayed alongside their corresponding class names to enable **human-interpretable inspection** of the vision module's behavior.

This analysis supports:

- verification of semantic controllability at scale (102 classes),
- identification of visually confused or entangled categories,

- and assessment of how well the perception module internalizes class-level structure in a natural image domain.

Such qualitative grounding is critical in robotics, where perception outputs must remain interpretable and diagnosable before downstream use in autonomy pipelines.

In [17]:
```
import torch
import matplotlib.pyplot as plt
import os

# Load trained Generator
G = Generator64().to(DEVICE)
G.load_state_dict(torch.load(os.path.join(save_dir, "generator.pth"), map_loca
G.eval()  # Set to evaluation mode

# Prepare Correct Class Names (Oxford-102 flowers)
class_names = [
    'pink primrose', 'hard-leaved pocket orchid', 'canterbury bells', 'sweet p
    'tiger lily', 'moon orchid', 'bird of paradise', 'monkshood', 'globe thist
    "colt's foot", 'king protea', 'spear thistle', 'yellow iris', 'globe-flowe
    'peruvian lily', 'balloon flower', 'giant white arum lily', 'fire lily', '
    'red ginger', 'grape hyacinth', 'corn poppy', 'prince of wales feathers',
    'sweet william', 'carnation', 'garden phlox', 'love in the mist', 'mexican
    'ruby-lipped cattleya', 'cape flower', 'great masterwort', 'siam tulip', '
    'daffodil', 'sword lily', 'poinsettia', 'bolero deep blue', 'wallflower',
    'oxeye daisy', 'common dandelion', 'petunia', 'wild pansy', 'primula', 'su
    'bishop of llandaff', 'gaura', 'geranium', 'orange dahlia', 'pink-yellow d
    'japanese anemone', 'black-eyed susan', 'silverbush', 'californian poppy',
    'bearded iris', 'windflower', 'tree poppy', 'gazania', 'azalea', 'water li
    'morning glory', 'passion flower', 'lotus', 'toad lily', 'anthurium', 'fra
    'columbine', 'desert-rose', 'tree mallow', 'magnolia', 'cyclamen ', 'water
    'bee balm', 'ball moss', 'foxglove', 'bougainvillea', 'camellia', 'mallow'
    'blanket flower', 'trumpet creeper', 'blackberry lily'
]

# Generate samples
fixed_z = torch.randn(y_dim, z_dim, device=DEVICE)  # 1 sample per class
fixed_labels = torch.arange(y_dim, device=DEVICE)
fixed_labels_onehot = make_one_hot(fixed_labels, y_dim)

with torch.no_grad():
    fake_imgs = G(fixed_z, fixed_labels_onehot)

# Show and map flowers
fig, axes = plt.subplots(8, 13, figsize=(18, 12))
axes = axes.flatten()

fake_imgs = (fake_imgs + 1) / 2  # Denormalize from [-1,1] to [0,1]

for idx, (img, ax) in enumerate(zip(fake_imgs, axes)):
```

```
    img = img.detach().cpu().permute(1, 2, 0)
    ax.imshow(img)
    if idx < len(class_names):
        ax.set_title(f"{class_names[idx]}", fontsize=6)
    ax.axis('off')

# Hide extra axes if any
for ax in axes[len(fake_imgs):]:
    ax.axis('off')

plt.tight_layout()
plt.show()
```

```
C:\Users\aymisxx\AppData\Local\Temp\ipykernel_10568\595977269.py:7: FutureWarni
ng: You are using `torch.load` with `weights_only=False` (the current default v
alue), which uses the default pickle module implicitly. It is possible to const
ruct malicious pickle data which will execute arbitrary code during unpickling
(See https://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models
for more details). In a future release, the default value for `weights_only` wi
ll be flipped to `True`. This limits the functions that could be executed durin
g unpickling. Arbitrary objects will no longer be allowed to be loaded via this
mode unless they are explicitly allowlisted by the user via `torch.serializatio
n.add_safe_globals`. We recommend you start setting `weights_only=True` for any
use case where you don't have full control of the loaded file. Please open an i
ssue on GitHub for any issues related to this experimental feature.
  G.load_state_dict(torch.load(os.path.join(save_dir, "generator.pth"), map_loc
ation=DEVICE))
```

Oxford-102 Flower Classes: Index → Semantic Label Mapping

Each class index represents a distinct semantic category used for label-conditioned robot vision.

1. Pink primrose
2. Hard-leaved pocket orchid
3. Canterbury bells
4. Sweet pea
5. English marigold
6. Tiger lily
7. Moon orchid
8. Bird of paradise
9. Monkshood
10. Globe thistle
11. Snapdragon
12. Colt’s foot
13. King protea
14. Spear thistle
15. Yellow iris
16. Globe-flower
17. Purple coneflower
18. Peruvian lily
19. Ball moss
20. Mexican petunia
21. Bromelia
22. Blanket flower
23. Trumpet creeper
24. Black-eyed susan
25. Pontederia
26. Bolero deep blue
27. Bougainvillea
28. Camellia
29. Mallow
30. Mexican aster
31. Alpine sea holly
32. Ruby-lipped cattleya
33. Cape flower
34. Great masterwort

35. Siam tulip
36. Lenten rose
37. Barbeton daisy
38. Daffodil
39. Sword lily
40. Poinsettia
41. Gaura
42. Geranium
43. Orange dahlia
44. Pink-yellow dahlia
45. Cautleya spicata
46. Japanese anemone
47. Blackberry lily
48. Tree poppy
49. Gazania
50. Azalea
51. Water lily
52. Rose
53. Thorn apple
54. Morning glory
55. Passion flower
56. Lotus
57. Toad lily
58. Anthurium
59. Frangipani
60. Clematis
61. Hibiscus
62. Columbine
63. Desert-rose
64. Tree mallow
65. Magnolia
66. Cyclamen
67. Watercress
68. Canna lily
69. Hippeastrum
70. Bee balm
71. Balloon flower
72. Oxeye daisy
73. Fire lily
74. Pincushion flower

75. Fritillary
76. Red ginger
77. Grape hyacinth
78. Corn poppy
79. Prince of wales feathers
80. Stemless gentian
81. Artichoke
82. Sweet william
83. Carnation
84. Garden phlox
85. Love in the mist
86. Mexican sunflower
87. Wild pansy
88. Primula
89. Sunflower
90. Pelargonium
91. Bishop of llandaff
92. Gaillardia
93. Buttercup
94. Oxlip
95. Tiger flower
96. Rose mallow
97. Snapdragon
98. Columbine
99. Colt’s foot
100. King protea
101. Spear thistle
102. Globe thistle

In [18]:

```
import os
import torch
import random
from PIL import Image
import numpy as np
import matplotlib.pyplot as plt
import scipy.io

# Paths
real_image_folder = os.path.join(oxford102_root, "jpg")
labels_mat_path = os.path.join(oxford102_root, "imagelabels.mat")

class_names = [
    'pink primrose', 'hard-leaved pocket orchid', 'canterbury bells', 'sweet p
    'tiger lily', 'moon orchid', 'bird of paradise', 'monkshood', 'globe thist
```

```
    "colt's foot", 'king protea', 'spear thistle', 'yellow iris', 'globe-flowe
    'peruvian lily', 'balloon flower', 'giant white arum lily', 'fire lily', '
    'red ginger', 'grape hyacinth', 'corn poppy', 'prince of wales feathers',
    'sweet william', 'carnation', 'garden phlox', 'love in the mist', 'mexican
    'ruby-lipped cattleya', 'cape flower', 'great masterwort', 'siam tulip', '
    'daffodil', 'sword lily', 'poinsettia', 'bolero deep blue', 'wallflower',
    'oxeye daisy', 'common dandelion', 'petunia', 'wild pansy', 'primula', 'su
    'bishop of llandaff', 'gaura', 'geranium', 'orange dahlia', 'pink-yellow d
    'japanese anemone', 'black-eyed susan', 'silverbush', 'californian poppy',
    'bearded iris', 'windflower', 'tree poppy', 'gazania', 'azalea', 'water li
    'morning glory', 'passion flower', 'lotus', 'toad lily', 'anthurium', 'fra
    'columbine', 'desert-rose', 'tree mallow', 'magnolia', 'cyclamen ', 'water
    'bee balm', 'ball moss', 'foxglove', 'bougainvillea', 'camellia', 'mallow'
    'blanket flower', 'trumpet creeper', 'blackberry lily'
]
```

Class-Specific Grounding: Real vs Generated Comparisons

This section performs **targeted semantic grounding checks** for selected flower classes in the Oxford-102 dataset.

This evaluation answers a critical question:

> *Given an explicit semantic prior, can the vision module generate a plausible visual observation consistent with real sensor data?*

For a small set of representative classes (Corn Poppy, English Marigold, Snapdragon), we:

- retrieve a **real image** from the dataset corresponding to the target class,
- generate a **synthetic image** using the trained label-conditioned generator,
- and visualize both side-by-side for direct comparison.

This class-specific analysis enables:

- fine-grained inspection of semantic alignment,
- identification of intra-class variability and background bias,
- and qualitative validation of label-conditioned robot vision behavior beyond aggregate metrics.

In [22]:
```
import os
import random
import torch
import matplotlib.pyplot as plt
```

```python
from PIL import Image
import scipy.io

# Target flower classes for inspection
selected_flower_names = ["corn poppy", "english marigold", "snapdragon"]

# Load labels once
labels_data = scipy.io.loadmat(labels_mat_path)
real_labels = labels_data["labels"][0] - 1  # MATLAB labels are 1-indexed

image_files = sorted(os.listdir(real_image_folder))

# Loop over selected classes
for selected_flower_name in selected_flower_names:
    selected_label = class_names.index(selected_flower_name)
    print(f"Selected Flower: {selected_flower_name} (Label {selected_label})")

    # Find matching real images
    matching_indices = [i for i, lbl in enumerate(real_labels) if lbl == selec
    real_idx = random.choice(matching_indices)
    real_img_path = os.path.join(real_image_folder, image_files[real_idx])

    # Load and preprocess real image
    real_img = Image.open(real_img_path).convert("RGB")
    real_img = real_img.resize((img_size, img_size))

    # Generate synthetic image
    z = torch.randn(1, z_dim, device=DEVICE)
    label_tensor = torch.tensor([selected_label], device=DEVICE)
    label_onehot = make_one_hot(label_tensor, y_dim)

    with torch.no_grad():
        fake_img = G(z, label_onehot)

    fake_img = (fake_img + 1) / 2  # Denormalize
    fake_img = fake_img.squeeze(0).permute(1, 2, 0).cpu().numpy()

    # Side-by-side visualization
    fig, axes = plt.subplots(1, 2, figsize=(8, 4))

    axes[0].imshow(real_img)
    axes[0].set_title(f"Real: {selected_flower_name}")
    axes[0].axis("off")

    axes[1].imshow(fake_img)
    axes[1].set_title(f"Generated: {selected_flower_name}")
    axes[1].axis("off")

    plt.tight_layout()
    plt.show()
```

```
Selected Flower: corn poppy (Label 25)
```

Real: corn poppy

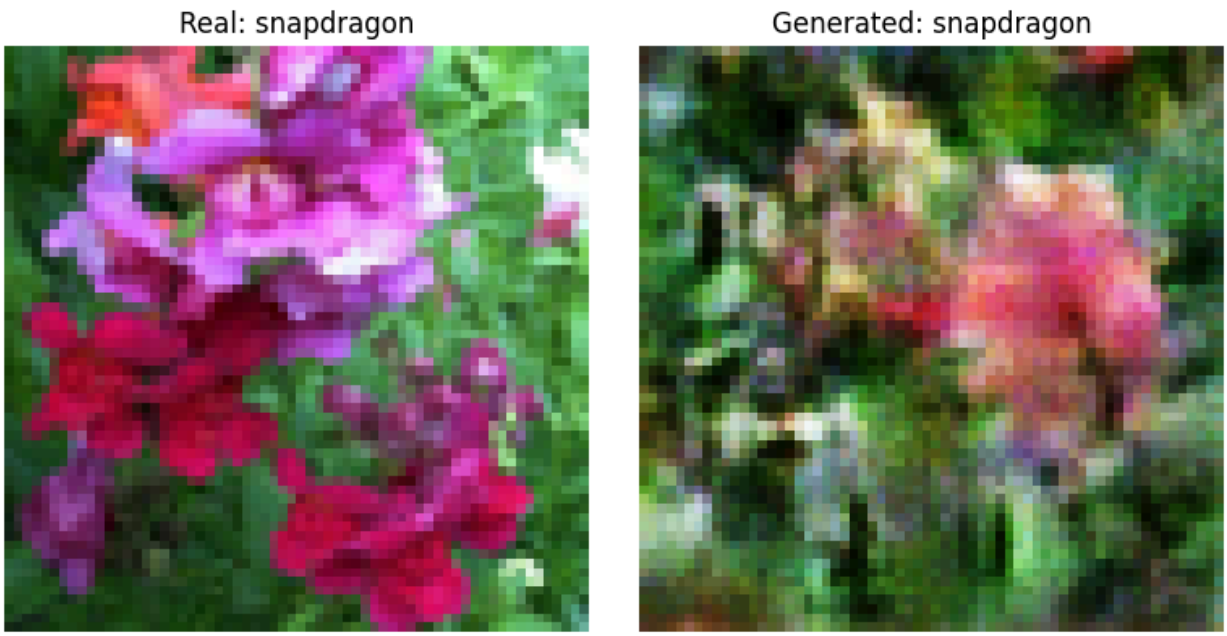
Generated: corn poppy

```
Selected Flower: english marigold (Label 4)
```

Real: english marigold

Generated: english marigold

```
Selected Flower: snapdragon (Label 10)
```



Class-Specific Grounding Results: Real vs Generated Comparisons

The figure above presents **side-by-side comparisons** between real dataset images and corresponding images generated by the label-conditioned robot vision module for three representative classes: **Corn Poppy**, **English Marigold**, and **Snapdragon**.

This visualization evaluates **semantic grounding under explicit class conditioning** rather than photorealism alone.

Observations

- **Semantic Color and Structure Preservation**
  For all three classes, the generated images capture dominant **class-specific color distributions** (e.g., red for corn poppy, yellow–orange for marigold, pink–purple for snapdragon), indicating that the generator has internalized coarse semantic cues associated with each label.

- **Loss of Fine-Grained Detail**
  Compared to real images, generated samples exhibit blurred edges and reduced structural clarity. This is expected given:

    - the fixed low resolution (64×64),
    - limited model capacity,
    - and the high intra-class variability of natural flower images.

- **Background Entanglement**
  Generated images show noticeable blending between foreground and background textures, reflecting a common failure mode in conditional generative vision systems when trained on complex natural scenes.

Robot Interpretation

In a robotics context, these results suggest that the vision module:

- successfully **respects semantic priors** (label → visual consistency),
- provides **coarse perceptual hypotheses** suitable for downstream reasoning or simulation,
- but is **not yet suitable for high-precision visual tasks** requiring sharp object boundaries or fine texture discrimination.

This qualitative grounding confirms that the system learns **class-level visual representations**, while also clearly exposing the perceptual limits imposed by resolution, architecture depth, and dataset complexity.

CUB-200

CUB-200 Configuration

This section extends the label-conditioned robot vision framework to the **CUB-200 (Caltech–UCSD Birds)** dataset, representing a **high-entropy, fine-grained perception challenge**.

The CUB-200 models scenarios where a vision system must distinguish between **visually similar object categories** that differ only in subtle attributes such as shape, texture, pose, or color patterns. This closely mirrors real robotic perception tasks involving:

- fine-grained object recognition (e.g., tools, components, debris),
- category differentiation under small inter-class variance,
- and perception under cluttered, natural backgrounds.

Compared to Oxford-102:

- The number of classes increases to **200**, significantly expanding the semantic space.
- Inter-class differences are more subtle, increasing the risk of **class**

entanglement.
- Visual cues are less dominated by color alone, requiring stronger structural and textural reasoning.

CUB-200 therefore serves as a **stress test** for label-conditioned robot vision, evaluating whether semantic conditioning remains effective when visual distinctions are nuanced and data complexity is high.

Key configuration decisions:

- **Higher image resolution (96×96 RGB)** compared to Oxford-102 (64×64) to retain more structural detail needed for fine-grained discrimination.
- **Same latent dimension (`z_dim = 100`)** to isolate the effect of increased semantic and visual complexity.
- **Reduced training horizon (`epochs = 150`)** to balance computational cost with the increased difficulty and model capacity required at this scale.

This configuration sets up the label-conditioned robot vision module to be evaluated under conditions analogous to real robotic perception problems involving **high category counts, low inter-class separability, and cluttered natural imagery**.

In [30]:
```
import os
import torch

dataset_name = "CUB200"

# Hyperparameters
batch_size  = 64
z_dim       = 100
y_dim       = 200          # 200 bird species
img_size    = 96
img_channels= 3
img_dim     = img_size * img_size * img_channels
lr          = 0.0002
epochs      = 150

save_dir = f"./outputs/{dataset_name}"
os.makedirs(save_dir, exist_ok=True)

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")

print(f"Using device: {DEVICE}")
```

```
Using device: cuda
```

CUB-200 Dataset Acquisition and Preparation

This cell handles the **download and extraction** of the CUB-200-2011 dataset, which serves as the final and most demanding visual benchmark in this project.

This step corresponds to **sensor data provisioning for a high-entropy environment**, where:

- visual observations are complex and fine-grained,
- semantic annotations are extensive (200 categories),
- and raw data must be organized before being passed into perception pipelines.

Key aspects of this setup:

- The dataset is retrieved from an authoritative academic source and stored locally for reproducibility.
- Automated checks prevent redundant downloads or re-extraction, mirroring best practices in robotics data pipelines.
- The extracted directory structure preserves original annotations and image organization required for fine-grained perception tasks.

This preparation step ensures that the CUB-200 dataset is **ready for downstream preprocessing, dataset interfacing, and training** of the label-conditioned robot vision module.

In [31]:
```
import urllib.request
import tarfile

cub_root = "./data/CUB_200_2011"
cub_tar_path = "./data/CUB_200_2011.tgz"
cub_url = "https://data.caltech.edu/records/65de6-vp158/files/CUB_200_2011.tgz

os.makedirs("./data", exist_ok=True)

# Download dataset if not present
if not os.path.exists(cub_tar_path):
    print("Downloading CUB-200-2011...")
    urllib.request.urlretrieve(cub_url, cub_tar_path)
    print("Downloaded!")

# Extract dataset if not already extracted
if not os.path.exists(cub_root):
    print("Extracting CUB-200-2011...")
    with tarfile.open(cub_tar_path, "r:gz") as tar:
        tar.extractall(path="./data")
    print("Extracted!")
```

```
print("CUB-200 dataset ready.")
```

```
Downloading CUB-200-2011...
Downloaded!
Extracting CUB-200-2011...
Extracted!
CUB-200 dataset ready.
```

CUB-200 Dataset Interface and Vision Preprocessing

This section defines the **custom dataset interface** for integrating the CUB-200-2011 bird dataset into the label-conditioned robot vision pipeline.

This step formalizes how **raw camera observations** and **fine-grained semantic annotations** are exposed to the perception module in a structured and reproducible manner.

Key design considerations:

- **High-resolution RGB preprocessing (96×96)** preserves subtle visual cues (plumage patterns, shape, pose) essential for fine-grained object discrimination.
- **Consistent normalization** ensures stable learning dynamics across datasets and aligns generator output ranges.
- **Explicit label parsing and zero-indexing** convert dataset annotations into a semantic format compatible with conditioning mechanisms.
- A custom `Dataset` abstraction mirrors robotics software practice, separating data access logic from model design and training.

The resulting DataLoader provides shuffled, batch-aligned image–label pairs, enabling scalable training and evaluation of label-conditioned robot vision under **high semantic cardinality and low inter-class separability**—conditions typical of real-world robotic perception tasks.

In [32]:
```
import scipy.io
from torchvision import transforms
from PIL import Image
from torch.utils.data import Dataset, DataLoader

# Define transforms
transform = transforms.Compose([
    transforms.Resize((img_size, img_size)),
    transforms.ToTensor(),
```

```
    transforms.Normalize([0.5, 0.5, 0.5], [0.5, 0.5, 0.5])
])

# Custom Dataset
class CUBDataset(Dataset):
    def __init__(self, root_dir, transform=None):
        self.root_dir = root_dir
        self.transform = transform

        # Read image paths
        with open(os.path.join(root_dir, "images.txt")) as f:
            self.image_paths = [line.strip().split()[1] for line in f.readline

        # Read labels
        with open(os.path.join(root_dir, "image_class_labels.txt")) as f:
            self.labels = [int(line.strip().split()[1]) - 1 for line in f.read

        assert len(self.image_paths) == len(self.labels), "Mismatch between im

    def __len__(self):
        return len(self.image_paths)

    def __getitem__(self, idx):
        img_path = os.path.join(self.root_dir, "images", self.image_paths[idx]
        image = Image.open(img_path).convert("RGB")
        label = self.labels[idx]

        if self.transform:
            image = self.transform(image)

        return image, label

# Instantiate
cub_dataset = CUBDataset(cub_root, transform=transform)
cub_loader = DataLoader(cub_dataset, batch_size=batch_size, shuffle=True, drop

print(f"CUB-200 Dataset ready: {len(cub_dataset)} images across {y_dim} classe
```

```
CUB-200 Dataset ready: 11788 images across 200 classes.
```

CUB-200 Vision Module: Convolutional Label-Conditioned Architecture (96×96)

This cell defines the **96×96 RGB** label-conditioned generator and discriminator used for the CUB-200 experiment—the highest-entropy vision setting in this project.

The CUB-200 models a fine-grained perception regime where categories are numerous (**200 classes**) and visually similar. To retain subtle cues (shape, texture, pose), the vision module operates at a higher spatial resolution than Oxford-102

and uses a deeper convolutional hierarchy.

Conditioning Mechanism: Multi-Scale Semantic Injection

To keep semantic control effective across a large class space, the label vector y is injected at **every spatial stage**:

- `expand_y(y, h, w)` tiles the one-hot class prior across the feature map
- the tiled semantic map is concatenated with the visual feature tensor before each conv layer

This mirrors robotics perception practice where **semantic context acts as a prior** that influences features at multiple levels (coarse structure early, fine details later).

Generator96: Semantic-to-Observation Model (96×96 RGB)

The generator produces a camera-like observation:

$$ G(z, y) \rightarrow x \in \mathbb{R}^{3 \times 96 \times 96} $$

Design highlights:

- The concatenated (z, y) is projected into a spatial seed (6×6×512), providing a richer starting representation than 4×4 in the 64×64 model.
- A sequence of transposed convolutions upsamples: 6→12→24→48→96
- Batch normalization and ReLU stabilize feature growth across scales.
- Final `tanh` output matches the normalized image range $[-1, 1]$, consistent with preprocessing.

Interpretation: a **label-conditioned observation generator** capable of expressing higher-frequency structure required for fine-grained categories.

Discriminator96: Semantic Consistency Verifier (Fine-Grained Regime)

The discriminator evaluates realism and semantic consistency:

$$ D(x, y) \rightarrow \text{logit} $$

Design highlights:

- The image is downsampled through four convolutional stages: 96→48→24→12→6
- Semantic priors are concatenated at each stage to enforce conditional verification.
- LeakyReLU activations support stable gradients under adversarial learning.
- The final linear layer outputs a single logit (intended for **BCEWithLogitsLoss**).

Interpretation: a verifier that checks whether an observation could plausibly belong to the claimed semantic class.

Robotics Lens: What This Architecture Tests

This 96×96 conditional design is a direct stress test of label-conditioned robot vision under:

- **large semantic space** (200 classes),
- **low inter-class separability** (fine-grained differences),
- **background/pose variation** common in real-world robotic sensing.

Expected failure modes at this stage include class entanglement, texture hallucination, and background leakage, each of which is informative for diagnosing limits of semantic conditioning in robot vision systems.

In [33]:
```
import torch
import torch.nn as nn

def expand_y(y, h, w):
    return y.view(y.size(0), y.size(1), 1, 1).expand(-1, -1, h, w)

class Generator96(nn.Module):
    def __init__(self):
        super().__init__()
        # project z+y → 6×6×512
        self.fc = nn.Sequential(
            nn.Linear(z_dim + y_dim, 6 * 6 * 512),
            nn.BatchNorm1d(6 * 6 * 512),
            nn.ReLU(True)
        )
        self.conv_blocks = nn.ModuleList([
            nn.ConvTranspose2d(512 + y_dim, 256, 4, 2, 1, bias=False),  # 6→12
            nn.ConvTranspose2d(256 + y_dim, 128, 4, 2, 1, bias=False),  # 12→2
            nn.ConvTranspose2d(128 + y_dim,  64, 4, 2, 1, bias=False),  # 24→4
```

```
            nn.ConvTranspose2d( 64 + y_dim,   3, 4, 2, 1, bias=False),  # 48→9
        ])
        self.bns = nn.ModuleList([
            nn.BatchNorm2d(256),
            nn.BatchNorm2d(128),
            nn.BatchNorm2d(64),
        ])

    def forward(self, z, y):
        # z: (B, z_dim), y: (B, y_dim)
        x = torch.cat([z, y], dim=1)
        x = self.fc(x).view(-1, 512, 6, 6)
        for i, conv in enumerate(self.conv_blocks):
            h, w = x.shape[2:]
            y_exp = expand_y(y, h, w)
            x = torch.cat([x, y_exp], dim=1)
            x = conv(x)
            if i < len(self.bns):
                x = self.bns[i](x)
                x = nn.ReLU(True)(x)
            else:
                x = torch.tanh(x)
        return x

class Discriminator96(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv_blocks = nn.ModuleList([
            nn.Conv2d(3 + y_dim,   64, 4, 2, 1, bias=True),  # 96→48
            nn.Conv2d(64 + y_dim, 128, 4, 2, 1, bias=True),  # 48→24
            nn.Conv2d(128 + y_dim,256, 4, 2, 1, bias=True),  # 24→12
            nn.Conv2d(256 + y_dim,512, 4, 2, 1, bias=True),  # 12→6
        ])
        self.fc = nn.Linear(6 * 6 * 512, 1)

    def forward(self, x, y):
        # x: (B, 3, 96,96), y: (B, y_dim)
        for conv in self.conv_blocks:
            h, w = x.shape[2:]
            y_exp = expand_y(y, h, w)
            x = torch.cat([x, y_exp], dim=1)
            x = conv(x)
            x = nn.LeakyReLU(0.2, inplace=True)(x)
        x = x.view(x.size(0), -1)
        return self.fc(x)
```

Optimization Setup and Learning Objective (CUB-200)

This cell initializes the **learning components** for the CUB-200 label-conditioned robot vision module.

This step defines how the perception subsystem **adapts its internal representation** when exposed to a highly fine-grained visual environment.

Key elements:

- The **Generator96** and **Discriminator96** together form a conditional perception model operating at higher spatial resolution.
- **Adam optimization** is used to handle the unstable dynamics typical of adversarial learning under large semantic cardinality.
- **Binary Cross-Entropy with Logits** provides a numerically stable objective for evaluating both visual realism and semantic consistency.

This configuration establishes the **control and learning rules** governing how the robot vision module updates its parameters during training on the most challenging dataset in the project.

In [34]:
```
import torch.optim as optim

G = Generator96().to(DEVICE)
D = Discriminator96().to(DEVICE)

G_optimizer = optim.Adam(G.parameters(), lr=lr, betas=(0.5, 0.999))
D_optimizer = optim.Adam(D.parameters(), lr=lr, betas=(0.5, 0.999))

criterion = nn.BCEWithLogitsLoss()

print("Success!!")
```

```
Success!!
```

Semantic Encoding and Visual Monitoring Utilities (CUB-200)

This cell defines **supporting utilities** used throughout training and evaluation of the CUB-200 label-conditioned robot vision module.

These functions provide essential infrastructure for **semantic conditioning** and **perceptual monitoring**:

- **One-hot label encoding** converts discrete class identifiers into explicit semantic vectors, enabling controlled interaction between symbolic priors and visual generation.
- **Image grid visualization** periodically records batches of generated observations, allowing qualitative inspection of:
    - semantic consistency across classes,

- visual diversity within batches,
- and training progression over epochs.

Such utilities are critical in robotics workflows, where perception systems must remain **diagnosable and interpretable**, especially when operating under high semantic cardinality and complex visual environments.

In [35]:
```
def make_one_hot(labels, num_classes):
    return torch.zeros(labels.size(0), num_classes, device=labels.device).scat

def save_image_grid(images, labels, epoch, save_dir, nrow=8):
    import matplotlib.pyplot as plt
    images = images.detach().cpu()
    images = (images + 1) / 2
    batch_size = images.size(0)
    fig, axes = plt.subplots(nrows=(batch_size // nrow) + 1, ncols=nrow, figsi
    axes = axes.flatten()

    for img, ax in zip(images, axes):
        img = img.permute(1, 2, 0)
        ax.imshow(img)
        ax.axis('off')

    for ax in axes[batch_size:]:
        ax.axis('off')

    plt.tight_layout()
    os.makedirs(save_dir, exist_ok=True)
    plt.savefig(os.path.join(save_dir, f"epoch_{epoch:03d}.png"))
    plt.close()
```

Offline Training Loop: CUB-200 Label-Conditioned Robot Vision (Fine-Grained Regime)

This section implements the **offline training procedure** for the CUB-200 label-conditioned robot vision module, representing the most challenging perception setting in this project.

This loop trains a vision subsystem under:

- **large semantic cardinality** (200 classes),
- **fine-grained inter-class differences**,
- and **natural-scene variability** (pose, background, lighting).

Training dynamics:

- The **Discriminator** learns to verify whether an observed image is both

realistic and *consistent with the provided semantic prior*.

- The **Generator** learns to synthesize camera-like RGB observations that satisfy the semantic constraint while remaining plausible under the dataset distribution.
- Alternating optimization steps reflect the adversarial nature of the learning objective and expose robotics-relevant failure modes such as class entanglement, texture hallucination, and background leakage.

Monitoring and diagnostics:

- A fixed set of latent vectors and class labels is used to produce periodic checkpoint grids, enabling consistent visual comparison over time.
- These qualitative checkpoints complement scalar losses and help identify whether semantic conditioning remains effective as dataset difficulty increases.

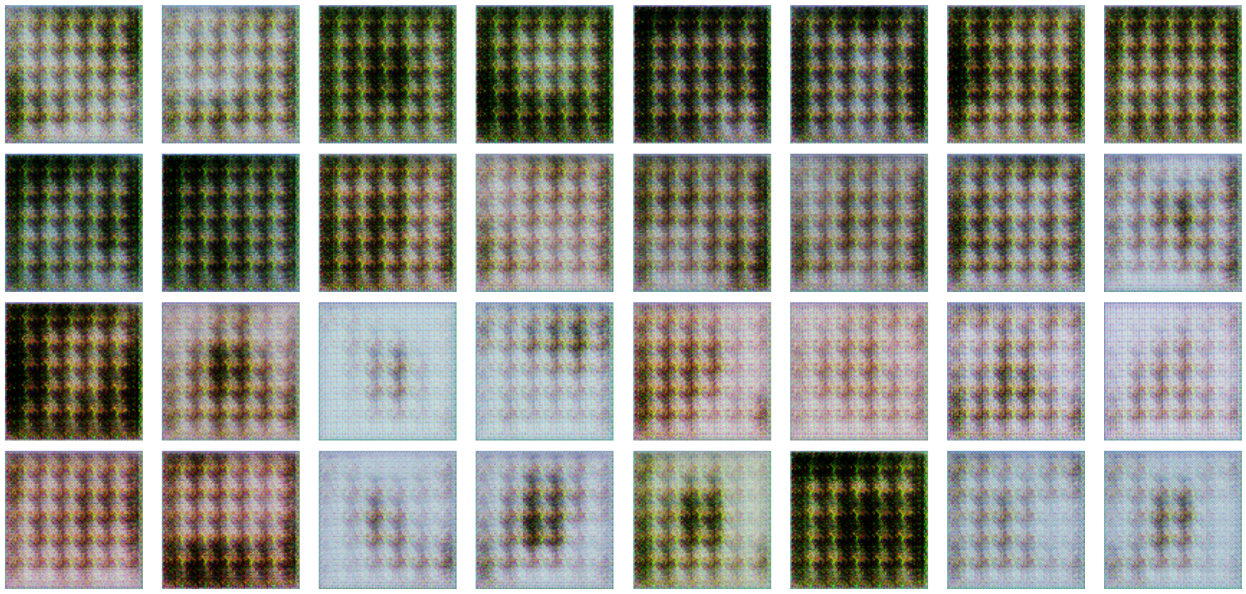
At completion, the model weights and loss histories are saved for reproducibility, post-hoc analysis, and cross-dataset comparison of training stability and perceptual outcomes.

In [36]:
```
import numpy as np
from tqdm import tqdm
import matplotlib.pyplot as plt

fixed_z = torch.randn(y_dim * 2, z_dim, device=DEVICE)
fixed_labels = torch.arange(y_dim, device=DEVICE).repeat_interleave(2)
fixed_labels_onehot = make_one_hot(fixed_labels, y_dim)

D_losses = []
G_losses = []

for epoch in range(1, epochs + 1):
    loop = tqdm(cub_loader, leave=False, desc=f"Epoch {epoch}/{epochs}")
    for real_imgs, labels in loop:
        real_imgs, labels = real_imgs.to(DEVICE), labels.to(DEVICE)
        bsize = real_imgs.size(0)

        real_targets = torch.ones(bsize, 1, device=DEVICE)
        fake_targets = torch.zeros(bsize, 1, device=DEVICE)

        labels_onehot = make_one_hot(labels, y_dim)

        D_optimizer.zero_grad()
        D_real = D(real_imgs, labels_onehot)
        D_loss_real = criterion(D_real, real_targets)

        z = torch.randn(bsize, z_dim, device=DEVICE)
        fake_imgs = G(z, labels_onehot)
```

```
        D_fake = D(fake_imgs.detach(), labels_onehot)
        D_loss_fake = criterion(D_fake, fake_targets)

        D_loss = D_loss_real + D_loss_fake
        D_loss.backward()
        D_optimizer.step()

        G_optimizer.zero_grad()
        z = torch.randn(bsize, z_dim, device=DEVICE)
        fake_imgs = G(z, labels_onehot)
        D_fake = D(fake_imgs, labels_onehot)
        G_loss = criterion(D_fake, real_targets)

        G_loss.backward()
        G_optimizer.step()

        D_losses.append(D_loss.item())
        G_losses.append(G_loss.item())

    print(f"Epoch [{epoch}/{epochs}]  D_loss: {D_loss.item():.4f}  G_loss: {G_

    if epoch % 30 == 0 or epoch == 1:
        with torch.no_grad():
            fake_imgs_fixed = G(fixed_z, fixed_labels_onehot)

        save_image_grid(fake_imgs_fixed, fixed_labels, epoch, save_dir)

        fig, axes = plt.subplots(4, 8, figsize=(16, 8))
        axes = axes.flatten()
        fake_imgs_fixed = fake_imgs_fixed.detach().cpu()
        fake_imgs_fixed = (fake_imgs_fixed + 1) / 2

        for img, ax in zip(fake_imgs_fixed, axes):
            img = img.permute(1, 2, 0)
            ax.imshow(img)
            ax.axis('off')

        for ax in axes[len(fake_imgs_fixed):]:
            ax.axis('off')

        plt.suptitle(f"Generated Samples at Epoch {epoch}", fontsize=16)
        plt.tight_layout()
        plt.show()

np.save(os.path.join(save_dir, "D_losses.npy"), np.array(D_losses))
np.save(os.path.join(save_dir, "G_losses.npy"), np.array(G_losses))
torch.save(G.state_dict(), os.path.join(save_dir, "generator.pth"))
torch.save(D.state_dict(), os.path.join(save_dir, "discriminator.pth"))

print("training done and models/losses saved.")
```

```
Epoch [1/150]  D_loss: 1.3002  G_loss: 0.9540
```

Generated Samples at Epoch 1



```
Epoch [2/150]  D_loss: 1.2962  G_loss: 0.7648

Epoch [3/150]  D_loss: 1.3199  G_loss: 0.7955

Epoch [4/150]  D_loss: 1.2901  G_loss: 0.8749

Epoch [5/150]  D_loss: 1.2580  G_loss: 0.9823

Epoch [6/150]  D_loss: 0.8043  G_loss: 2.0081

Epoch [7/150]  D_loss: 0.7378  G_loss: 2.1316

Epoch [8/150]  D_loss: 1.0296  G_loss: 1.6976

Epoch [9/150]  D_loss: 1.1249  G_loss: 1.6384

Epoch [10/150]  D_loss: 0.9738  G_loss: 2.2198

Epoch [11/150]  D_loss: 1.2052  G_loss: 1.5880

Epoch [12/150]  D_loss: 1.1253  G_loss: 1.3370

Epoch [13/150]  D_loss: 0.8328  G_loss: 2.1570

Epoch [14/150]  D_loss: 0.9884  G_loss: 2.4092

Epoch [15/150]  D_loss: 0.9106  G_loss: 2.1456

Epoch [16/150]  D_loss: 0.6301  G_loss: 2.9808
```

```
Epoch [17/150]  D_loss: 0.8354  G_loss: 2.6797

Epoch [18/150]  D_loss: 0.8891  G_loss: 2.4513

Epoch [19/150]  D_loss: 0.7837  G_loss: 2.5612

Epoch [20/150]  D_loss: 0.7403  G_loss: 2.0210

Epoch [21/150]  D_loss: 1.1029  G_loss: 1.9946

Epoch [22/150]  D_loss: 0.8655  G_loss: 2.6906

Epoch [23/150]  D_loss: 0.6496  G_loss: 2.3629

Epoch [24/150]  D_loss: 0.8982  G_loss: 2.1668

Epoch [25/150]  D_loss: 0.6510  G_loss: 2.1911

Epoch [26/150]  D_loss: 0.6645  G_loss: 2.3661

Epoch [27/150]  D_loss: 0.6716  G_loss: 2.4114

Epoch [28/150]  D_loss: 0.7819  G_loss: 2.6753

Epoch [29/150]  D_loss: 0.7072  G_loss: 2.0910

Epoch [30/150]  D_loss: 0.8236  G_loss: 2.0279
```

Generated Samples at Epoch 30

```
Epoch [31/150]  D_loss: 0.7821  G_loss: 2.6582

Epoch [32/150]  D_loss: 0.6122  G_loss: 2.5015
```

```
Epoch [33/150]  D_loss: 0.7303  G_loss: 2.1651
Epoch [34/150]  D_loss: 0.5889  G_loss: 2.3994
Epoch [35/150]  D_loss: 0.6738  G_loss: 2.2642
Epoch [36/150]  D_loss: 0.5985  G_loss: 2.4099
Epoch [37/150]  D_loss: 0.4644  G_loss: 2.2715
Epoch [38/150]  D_loss: 0.6143  G_loss: 3.4459
Epoch [39/150]  D_loss: 0.9220  G_loss: 3.8562
Epoch [40/150]  D_loss: 0.7399  G_loss: 2.2075
Epoch [41/150]  D_loss: 0.6634  G_loss: 2.6365
Epoch [42/150]  D_loss: 0.5869  G_loss: 3.0369
Epoch [43/150]  D_loss: 0.4532  G_loss: 3.1155
Epoch [44/150]  D_loss: 0.6822  G_loss: 2.8319
Epoch [45/150]  D_loss: 0.5984  G_loss: 2.3110
Epoch [46/150]  D_loss: 0.5488  G_loss: 3.0312
Epoch [47/150]  D_loss: 0.5652  G_loss: 2.5678
Epoch [48/150]  D_loss: 0.6070  G_loss: 2.9272
Epoch [49/150]  D_loss: 0.4886  G_loss: 3.2713
Epoch [50/150]  D_loss: 0.6895  G_loss: 3.6505
Epoch [51/150]  D_loss: 0.7592  G_loss: 3.2907
Epoch [52/150]  D_loss: 0.7611  G_loss: 3.4084
Epoch [53/150]  D_loss: 0.6230  G_loss: 3.8716
Epoch [54/150]  D_loss: 0.5936  G_loss: 3.4863
Epoch [55/150]  D_loss: 0.4190  G_loss: 3.1518
```

```
Epoch [56/150]  D_loss: 0.5269  G_loss: 2.8016

Epoch [57/150]  D_loss: 0.4419  G_loss: 2.9556

Epoch [58/150]  D_loss: 0.2532  G_loss: 3.5635

Epoch [59/150]  D_loss: 0.6619  G_loss: 2.5947

Epoch [60/150]  D_loss: 0.5116  G_loss: 3.0860
```

Generated Samples at Epoch 60

```
Epoch [61/150]  D_loss: 0.4590  G_loss: 3.3122

Epoch [62/150]  D_loss: 0.6101  G_loss: 3.4425

Epoch [63/150]  D_loss: 0.5533  G_loss: 3.7116

Epoch [64/150]  D_loss: 0.4446  G_loss: 3.3987

Epoch [65/150]  D_loss: 0.3855  G_loss: 3.9652

Epoch [66/150]  D_loss: 0.4743  G_loss: 4.3573

Epoch [67/150]  D_loss: 0.3846  G_loss: 3.4843

Epoch [68/150]  D_loss: 0.4900  G_loss: 3.2853

Epoch [69/150]  D_loss: 0.4501  G_loss: 3.6141

Epoch [70/150]  D_loss: 0.3457  G_loss: 4.3050

Epoch [71/150]  D_loss: 0.4512  G_loss: 4.3863
```

```
Epoch [72/150]  D_loss: 0.4101  G_loss: 3.9470
Epoch [73/150]  D_loss: 0.3950  G_loss: 3.1808
Epoch [74/150]  D_loss: 0.4241  G_loss: 4.1786
Epoch [75/150]  D_loss: 0.4701  G_loss: 3.2106
Epoch [76/150]  D_loss: 0.4647  G_loss: 2.9148
Epoch [77/150]  D_loss: 0.4600  G_loss: 3.6171
Epoch [78/150]  D_loss: 0.4553  G_loss: 4.4167
Epoch [79/150]  D_loss: 0.3532  G_loss: 4.7684
Epoch [80/150]  D_loss: 0.3598  G_loss: 3.2564
Epoch [81/150]  D_loss: 0.3543  G_loss: 3.8731
Epoch [82/150]  D_loss: 0.3454  G_loss: 3.8059
Epoch [83/150]  D_loss: 0.4506  G_loss: 4.4133
Epoch [84/150]  D_loss: 0.2181  G_loss: 4.2196
Epoch [85/150]  D_loss: 0.3296  G_loss: 4.2280
Epoch [86/150]  D_loss: 0.4155  G_loss: 4.5766
Epoch [87/150]  D_loss: 0.3887  G_loss: 5.2287
Epoch [88/150]  D_loss: 0.2448  G_loss: 4.4059
Epoch [89/150]  D_loss: 0.2895  G_loss: 5.3763
Epoch [90/150]  D_loss: 0.2492  G_loss: 4.3802
```

Generated Samples at Epoch 90

```
Epoch [91/150]  D_loss: 0.2263  G_loss: 3.9981

Epoch [92/150]  D_loss: 0.2566  G_loss: 5.4241

Epoch [93/150]  D_loss: 0.3553  G_loss: 4.5780

Epoch [94/150]  D_loss: 0.3315  G_loss: 4.5557

Epoch [95/150]  D_loss: 0.5034  G_loss: 5.3065

Epoch [96/150]  D_loss: 0.1680  G_loss: 5.0037

Epoch [97/150]  D_loss: 0.2099  G_loss: 4.4860

Epoch [98/150]  D_loss: 0.2482  G_loss: 4.6069

Epoch [99/150]  D_loss: 0.3706  G_loss: 4.2161

Epoch [100/150]  D_loss: 0.2247  G_loss: 5.0662

Epoch [101/150]  D_loss: 0.1982  G_loss: 5.3325

Epoch [102/150]  D_loss: 0.2723  G_loss: 4.9308

Epoch [103/150]  D_loss: 0.2991  G_loss: 3.6907

Epoch [104/150]  D_loss: 0.3662  G_loss: 6.0983

Epoch [105/150]  D_loss: 0.3387  G_loss: 5.5971
```

```
Epoch [106/150]  D_loss: 0.4255  G_loss: 6.1672

Epoch [107/150]  D_loss: 0.2119  G_loss: 5.8649

Epoch [108/150]  D_loss: 0.3316  G_loss: 5.7037

Epoch [109/150]  D_loss: 0.2438  G_loss: 5.5680

Epoch [110/150]  D_loss: 0.2940  G_loss: 6.4749

Epoch [111/150]  D_loss: 0.3698  G_loss: 5.6148

Epoch [112/150]  D_loss: 0.3172  G_loss: 5.6295

Epoch [113/150]  D_loss: 0.2877  G_loss: 5.6053

Epoch [114/150]  D_loss: 0.1577  G_loss: 5.0417

Epoch [115/150]  D_loss: 0.3304  G_loss: 5.5874

Epoch [116/150]  D_loss: 0.1710  G_loss: 5.2904

Epoch [117/150]  D_loss: 0.2850  G_loss: 5.1101

Epoch [118/150]  D_loss: 0.1747  G_loss: 5.4352

Epoch [119/150]  D_loss: 0.2469  G_loss: 5.4802

Epoch [120/150]  D_loss: 0.1841  G_loss: 6.1290
```

Generated Samples at Epoch 120

```
Epoch [121/150]  D_loss: 0.0891  G_loss: 4.1882
```

```
Epoch [122/150]  D_loss: 0.1932  G_loss: 5.0000
Epoch [123/150]  D_loss: 0.1680  G_loss: 5.6764
Epoch [124/150]  D_loss: 0.1800  G_loss: 6.9698
Epoch [125/150]  D_loss: 0.4042  G_loss: 6.0387
Epoch [126/150]  D_loss: 0.2010  G_loss: 5.8791
Epoch [127/150]  D_loss: 0.1866  G_loss: 6.3689
Epoch [128/150]  D_loss: 0.2086  G_loss: 5.3929
Epoch [129/150]  D_loss: 0.1561  G_loss: 5.7704
Epoch [130/150]  D_loss: 0.2969  G_loss: 5.0027
Epoch [131/150]  D_loss: 0.0952  G_loss: 4.9594
Epoch [132/150]  D_loss: 0.2672  G_loss: 5.5324
Epoch [133/150]  D_loss: 0.2984  G_loss: 6.7135
Epoch [134/150]  D_loss: 0.2496  G_loss: 6.1241
Epoch [135/150]  D_loss: 0.1521  G_loss: 6.9788
Epoch [136/150]  D_loss: 0.1932  G_loss: 7.2440
Epoch [137/150]  D_loss: 0.1915  G_loss: 7.2295
Epoch [138/150]  D_loss: 0.2511  G_loss: 5.5407
Epoch [139/150]  D_loss: 0.1300  G_loss: 5.8520
Epoch [140/150]  D_loss: 0.1925  G_loss: 6.1075
Epoch [141/150]  D_loss: 0.1582  G_loss: 7.1472
Epoch [142/150]  D_loss: 0.3224  G_loss: 6.5104
Epoch [143/150]  D_loss: 0.1229  G_loss: 6.1146
Epoch [144/150]  D_loss: 0.1833  G_loss: 5.2083
```

```
Epoch [145/150]  D_loss: 0.4713  G_loss: 9.4688

Epoch [146/150]  D_loss: 0.2502  G_loss: 6.2558

Epoch [147/150]  D_loss: 0.2544  G_loss: 5.9912

Epoch [148/150]  D_loss: 0.3101  G_loss: 7.0077

Epoch [149/150]  D_loss: 0.1866  G_loss: 6.6880

Epoch [150/150]  D_loss: 0.1939  G_loss: 5.9857
```

```
training done and models/losses saved.
```

Class-Wise Generation: Semantic Coverage Across CUB-200

This section performs a **class-wise qualitative evaluation** of the trained CUB-200 label-conditioned robot vision module by generating **one synthetic bird image per semantic class**.

This evaluates **semantic coverage** rather than visual fidelity. The goal is to verify whether the perception module can produce a **distinct visual hypothesis for every class** in a large and fine-grained semantic space (200 bird species).

What This Evaluation Tests

- **Semantic reachability**: whether each label in the 200-class space maps to a non-degenerate visual output.

- **Class isolation**: whether generated samples vary meaningfully across labels instead of collapsing to a few modes.
- **Scalability of conditioning**: whether explicit semantic priors remain effective as class count increases.

Interpretation of Results

- The generated images exhibit significant variability in color, texture, and structure across classes, indicating that the generator has internalized **class-level visual cues**.
- Visual sharpness and anatomical correctness are limited, which is expected given:
    - the fixed 96×96 resolution,
    - the complexity of fine-grained bird categories,
    - and the absence of explicit spatial or part-based supervision.
- Some classes may appear visually similar or noisy, highlighting common failure modes such as **class entanglement** and **background leakage**.

Relevance

In robotic perception pipelines, such class-wise visualization serves as a **diagnostic tool**:

- to confirm that semantic conditioning spans the full label space,
- to identify categories where perception is unreliable,
- and to guide architectural or data-centric improvements.

This experiment demonstrates that the vision module maintains **semantic controllability at scale**, even under the most challenging dataset considered in this project.

In [37]:
```
import torch
import matplotlib.pyplot as plt
import math

# Load the trained Generator
G = Generator96().to(DEVICE)
G.load_state_dict(torch.load(os.path.join(save_dir, "generator.pth")))
G.eval()

# Load class names
cub_classes = []
```

```
with open(os.path.join(cub_root, "classes.txt")) as f:
    for line in f:
        cub_classes.append(line.strip().split(" ", 1)[1])

# Generate one bird per class
def generate_bird_for_each_class():
    z = torch.randn(y_dim, z_dim, device=DEVICE)  # 200 random noise vectors
    labels = torch.arange(y_dim, device=DEVICE)   # Labels 0 to 199
    labels_onehot = make_one_hot(labels, y_dim)

    with torch.no_grad():
        fake_imgs = G(z, labels_onehot)

    fake_imgs = (fake_imgs + 1) / 2  # Denormalize
    fake_imgs = fake_imgs.detach().cpu()

    # Plotting
    ncols = 10
    nrows = math.ceil(y_dim / ncols)
    fig, axes = plt.subplots(nrows, ncols, figsize=(ncols * 2, nrows * 2))
    axes = axes.flatten()

    for img, label, ax in zip(fake_imgs, labels, axes):
        img = img.permute(1, 2, 0)
        ax.imshow(img)
        ax.set_title(cub_classes[label], fontsize=5)
        ax.axis('off')

    for ax in axes[len(fake_imgs):]:
        ax.axis('off')

    plt.tight_layout()
    plt.show()

# Call the function
generate_bird_for_each_class()
```

```
C:\Users\aymisxx\AppData\Local\Temp\ipykernel_10568\390559494.py:7: FutureWarni
ng: You are using `torch.load` with `weights_only=False` (the current default v
alue), which uses the default pickle module implicitly. It is possible to const
ruct malicious pickle data which will execute arbitrary code during unpickling
(See https://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models
for more details). In a future release, the default value for `weights_only` wi
ll be flipped to `True`. This limits the functions that could be executed durin
g unpickling. Arbitrary objects will no longer be allowed to be loaded via this
mode unless they are explicitly allowlisted by the user via `torch.serializatio
n.add_safe_globals`. We recommend you start setting `weights_only=True` for any
use case where you don't have full control of the loaded file. Please open an i
ssue on GitHub for any issues related to this experimental feature.
  G.load_state_dict(torch.load(os.path.join(save_dir, "generator.pth")))
```

001.Black_footed_Albatross
002.Laysan_Albatross
003.Sooty_Albatross
004.Groove_billed_Ani
005.Crested_Auklet
006.Least_Auklet
007.Parakeet_Auklet
008.Rhinoceros_Auklet
009.Brewer_Blackbird
010.Red_winged_Blackbird
011.Rusty_Blackbird
012.Yellow_headed_Blackbird
013.Bobolink
014.Indigo_Bunting
015.Lazuli_Bunting
016.Painted_Bunting
017.Cardinal
018.Spotted_Catbird
019.Gray_Catbird
020.Yellow_breasted_Chat
021.Eastern_Towhee
022.Chuck_will_Widow
023.Brandt_Cormorant
024.Red_faced_Cormorant
025.Pelagic_Cormorant
026.Bronzed_Cowbird
027.Shiny_Cowbird
028.Brown_Creeper
029.American_Crow
030.Fish_Crow
031.Black_billed_Cuckoo
032.Mangrove_Cuckoo
033.Yellow_billed_Cuckoo
034.Gray_crowned_Rosy_Finch
035.Purple_Finch
036.Northern_Flicker
037.Acadian_Flycatcher
038.Great_Crested_Flycatcher
039.Least_Flycatcher
040.Olive_sided_Flycatcher
041.Scissor_tailed_Flycatcher
042.Vermilion_Flycatcher
043.Yellow_bellied_Flycatcher
044.Frigatebird
045.Northern_Fulmar
046.Gadwall
047.American_Goldfinch
048.European_Goldfinch
049.Boat_tailed_Grackle
050.Eared_Grebe
051.Horned_Grebe
052.Pied_billed_Grebe
053.Western_Grebe
054.Blue_Grosbeak
055.Evening_Grosbeak
056.Pine_Grosbeak
057.Rose_breasted_Grosbeak
058.Pigeon_Guillemot
059.California_Gull
060.Glaucous_winged_Gull
061.Heermann_Gull
062.Herring_Gull
063.Ivory_Gull
064.Ring_billed_Gull
065.Slaty_backed_Gull
066.Western_Gull
067.Anna_Hummingbird
068.Ruby_throated_Hummingbird
069.Rufous_Hummingbird
070.Green_Violetear
071.Long_tailed_Jaeger
072.Pomarine_Jaeger
073.Blue_Jay
074.Florida_Jay
075.Green_Jay
076.Dark_eyed_Junco
077.Tropical_Kingbird
078.Gray_Kingbird
079.Belted_Kingfisher
080.Green_Kingfisher
081.Pied_Kingfisher
082.Ringed_Kingfisher
083.White_breasted_Kingfisher
084.Red_legged_Kittiwake
085.Horned_Lark
086.Pacific_Loon
087.Mallard
088.Western_Meadowlark
089.Hooded_Merganser
090.Red_breasted_Merganser
091.Mockingbird
092.Nighthawk
093.Clark_Nutcracker
094.White_breasted_Nuthatch
095.Baltimore_Oriole
096.Hooded_Oriole
097.Orchard_Oriole
098.Scott_Oriole
099.Ovenbird
100.Brown_Pelican
101.White_Pelican
102.Western_Wood_Pewee
103.Sayornis
104.American_Pipit
105.Whip_poor_Will
106.Horned_Puffin
107.Common_Raven
108.White_necked_Raven
109.American_Redstart
110.Geococcyx
111.Loggerhead_Shrike
112.Great_Grey_Shrike
113.Baird_Sparrow
114.Black_throated_Sparrow
115.Brewer_Sparrow
116.Chipping_Sparrow
117.Clay_colored_Sparrow
118.House_Sparrow
119.Field_Sparrow
120.Fox_Sparrow
121.Grasshopper_Sparrow
122.Harris_Sparrow
123.Henslow_Sparrow
124.Le_Conte_Sparrow
125.Lincoln_Sparrow
126.Nelson_Sharp_tailed_Sparrow
127.Savannah_Sparrow
128.Seaside_Sparrow
129.Song_Sparrow
130.Tree_Sparrow
131.Vesper_Sparrow
132.White_crowned_Sparrow
133.White_throated_Sparrow
134.Cape_Glossy_Starling
135.Bank_Swallow
136.Barn_Swallow
137.Cliff_Swallow
138.Tree_Swallow
139.Scarlet_Tanager
140.Summer_Tanager
141.Artic_Tern
142.Black_Tern
143.Caspian_Tern
144.Common_Tern
145.Elegant_Tern
146.Forsters_Tern
147.Least_Tern
148.Green_tailed_Towhee
149.Brown_Thrasher
150.Sage_Thrasher

CUB-200 Bird Classes: Index → Semantic Label Mapping

Each index corresponds to a unique semantic class used for label-conditioned robot vision.

1. Black_Footed_Albatross
2. Laysan_Albatross
3. Sooty_Albatross
4. Groove_Billed_Ani
5. Crested_Auklet
6. Least_Auklet
7. Parakeet_Auklet
8. Rhinoceros_Auklet
9. Brewer_Blackbird
10. Red_Winged_Blackbird
11. Rusty_Blackbird
12. Yellow_Headed_Blackbird
13. Bobolink
14. Indigo_Bunting
15. Lazuli_Bunting
16. Painted_Bunting
17. Cardinal
18. Spotted_Catbird
19. Gray_Catbird
20. Yellow_Breasted_Chat
21. Eastern_Towhee
22. Chuck_Will_Widow
23. Brandt_Cormorant
24. Red_Faced_Cormorant
25. Pelagic_Cormorant
26. Bronzed_Cowbird
27. Shiny_Cowbird
28. Brown_Creeper
29. American_Crow
30. Fish_Crow
31. Black_Billed_Cuckoo
32. Mangrove_Cuckoo
33. Yellow_Billed_Cuckoo
34. Gray_Crowned_Rosy_Finch

35. Purple_Finch
36. Northern_Flicker
37. Acadian_Flycatcher
38. Great_Crested_Flycatcher
39. Least_Flycatcher
40. Olive_Sided_Flycatcher
41. Scissor_Tailed_Flycatcher
42. Vermilion_Flycatcher
43. Yellow_Bellied_Flycatcher
44. Frigatebird
45. Northern_Fulmar
46. Gadwall
47. American_Goldfinch
48. European_Goldfinch
49. Boat_Tailed_Grackle
50. Eared_Grebe
51. Horned_Grebe
52. Pied_Billed_Grebe
53. Western_Grebe
54. Blue_Grosbeak
55. Evening_Grosbeak
56. Pine_Grosbeak
57. Rose_Breasted_Grosbeak
58. Pigeon_Guillemot
59. California_Gull
60. Glaucous_Winged_Gull
61. Heermann_Gull
62. Herring_Gull
63. Ivory_Gull
64. Ring_Billed_Gull
65. Slaty_Backed_Gull
66. Western_Gull
67. Anna_Hummingbird
68. Ruby_Throated_Hummingbird
69. Rufous_Hummingbird
70. Green_Violetear
71. Long_Tailed_Jaeger
72. Pomarine_Jaeger
73. Blue_Jay
74. Florida_Jay

75. Green_Jay
76. Dark_Eyed_Junco
77. Northern_Jacana
78. Green_Kingfisher
79. Pied_Kingfisher
80. Ringed_Kingfisher
81. Belted_Kingfisher
82. White_Breasted_Kingfisher
83. Red_Legged_Kittiwake
84. Horned_Lark
85. Pacific_Loon
86. Mallard
87. Western_Meadowlark
88. Hooded_Merganser
89. Red_Breasted_Merganser
90. Mockingbird
91. Nighthawk
92. Clark_Nutcracker
93. White_Breasted_Nuthatch
94. Baltimore_Oriole
95. Hooded_Oriole
96. Orchard_Oriole
97. Scott_Oriole
98. Ovenbird
99. Brown_Pelican
100. White_Pelican
101. Western_Wood_Pewee
102. Sayornis
103. American_Pipit
104. Whip_Poor_Will
105. Horned_Puffin
106. Common_Raven
107. White_Necked_Raven
108. American_Redstart
109. Geococcyx
110. Loggerhead_Shrike
111. Great_Grey_Shrike
112. Baird_Sparrow
113. Black_Throated_Sparrow
114. Brewer_Sparrow

115. Chipping_Sparrow
116. Clay_Colored_Sparrow
117. House_Sparrow
118. Field_Sparrow
119. Fox_Sparrow
120. Grasshopper_Sparrow
121. Harris_Sparrow
122. Henslow_Sparrow
123. Le_Conte_Sparrow
124. Lincoln_Sparrow
125. Nelson_Sharp_Tailed_Sparrow
126. Savannah_Sparrow
127. Seaside_Sparrow
128. Song_Sparrow
129. Tree_Sparrow
130. Vesper_Sparrow
131. White_Crowned_Sparrow
132. White_Throated_Sparrow
133. Cape_Glossy_Starling
134. Bank_Swallow
135. Barn_Swallow
136. Cliff_Swallow
137. Tree_Swallow
138. Scarlet_Tanager
139. Summer_Tanager
140. Artic_Tern
141. Black_Tern
142. Caspian_Tern
143. Common_Tern
144. Elegant_Tern
145. Forsters_Tern
146. Least_Tern
147. Green_Tailed_Towhee
148. Brown_Thrasher
149. Sage_Thrasher
150. Black_Capped_Vireo
151. Blue_Headed_Vireo
152. Philadelphia_Vireo
153. Red_Eyed_Vireo
154. Warbling_Vireo

155. White_Eyed_Vireo
156. Yellow_Throated_Vireo
157. Bay_Breasted_Warbler
158. Black_And_White_Warbler
159. Black_Throated_Blue_Warbler
160. Blue_Winged_Warbler
161. Canada_Warbler
162. Cape_May_Warbler
163. Cerulean_Warbler
164. Chestnut_Sided_Warbler
165. Golden_Winged_Warbler
166. Hooded_Warbler
167. Kentucky_Warbler
168. Magnolia_Warbler
169. Mourning_Warbler
170. Myrtle_Warbler
171. Nashville_Warbler
172. Orange_Crowned_Warbler
173. Palm_Warbler
174. Pine_Warbler
175. Prairie_Warbler
176. Prothonotary_Warbler
177. Swainson_Warbler
178. Tennessee_Warbler
179. Wilson_Warbler
180. Worm_Eating_Warbler
181. Yellow_Warbler
182. Northern_Waterthrush
183. Louisiana_Waterthrush
184. Bohemian_Waxwing
185. Cedar_Waxwing
186. American_Three_Toed_Woodpecker
187. Downy_Woodpecker
188. Hairy_Woodpecker
189. Red_Bellied_Woodpecker
190. Red_Cockaded_Woodpecker
191. Pileated_Woodpecker
192. Red_Headed_Woodpecker
193. White_Breasted_Woodpecker
194. American_Three_Toed_Woodpecker

195. Bewick_Wren
196. Cactus_Wren
197. Carolina_Wren
198. House_Wren
199. Marsh_Wren
200. Rock_Wren

We will be entering (target label = (corresponding number - 1)) for the bird we want to generate.

Class-Specific Grounding: Real vs Generated Comparisons (CUB-200)

This section performs **targeted semantic grounding checks** on the CUB-200 dataset using a small set of representative bird classes.

This evaluates whether the perception module can:

- respect a **specific semantic prior** (class label),
- generate a plausible visual hypothesis consistent with that label,
- and maintain class-level separation in a **fine-grained, high-cardinality setting**.

For each selected class, we:

1. randomly sample a **real image** from the dataset,
2. generate a **synthetic image** conditioned on the same class label,
3. visualize both **side-by-side** for direct qualitative comparison.

This type of class-specific inspection is critical in robotics, where perception systems must be **interpretable and diagnosable** at the level of individual semantic categories rather than only aggregate statistics.

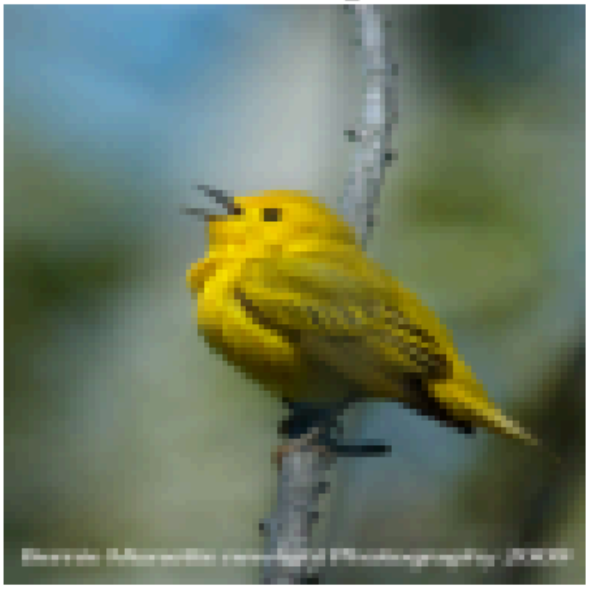
In [38]:
```
import torch
import matplotlib.pyplot as plt
import random
from PIL import Image
import os

# Load trained Generator
G = Generator96().to(DEVICE)
G.load_state_dict(torch.load(os.path.join(save_dir, "generator.pth"), map_loca
G.eval()

# Load class names
```

```python
with open(os.path.join(cub_root, "classes.txt")) as f:
    class_names = [line.strip().split(" ", 1)[1] for line in f.readlines()]

# Selected examples: (label, name for clarity)
target_examples = [
    (181, "Yellow Warbler"),
    (160, "Blue-winged Warbler"),
    (113, "Black-throated Sparrow"),
    (73,  "Florida Jay"),
]

fig, axes = plt.subplots(len(target_examples), 2, figsize=(8, 4 * len(target_e

for row, (target_label, _) in enumerate(target_examples):

    # Real image
    real_indices = [i for i, lbl in enumerate(cub_dataset.labels) if lbl == ta
    real_idx = random.choice(real_indices)
    real_img_path = os.path.join(cub_root, "images", cub_dataset.image_paths[r

    real_img = Image.open(real_img_path).convert("RGB")
    real_img = transform(real_img)
    real_img = (real_img + 1) / 2
    real_img = real_img.permute(1, 2, 0).cpu()

    # Generated image
    z = torch.randn(1, z_dim, device=DEVICE)
    label_tensor = torch.tensor([target_label], device=DEVICE)
    label_onehot = make_one_hot(label_tensor, y_dim)

    with torch.no_grad():
        fake_img = G(z, label_onehot)

    fake_img = (fake_img + 1) / 2
    fake_img = fake_img.squeeze(0).permute(1, 2, 0).cpu()

    # Plot
    axes[row, 0].imshow(real_img)
    axes[row, 0].set_title(f"Real: {class_names[target_label]}")
    axes[row, 0].axis("off")

    axes[row, 1].imshow(fake_img)
    axes[row, 1].set_title(f"Generated: {class_names[target_label]}")
    axes[row, 1].axis("off")

plt.suptitle("CUB-200: Class-Specific Real vs Generated Comparisons", fontsize
plt.tight_layout()
plt.show()
```

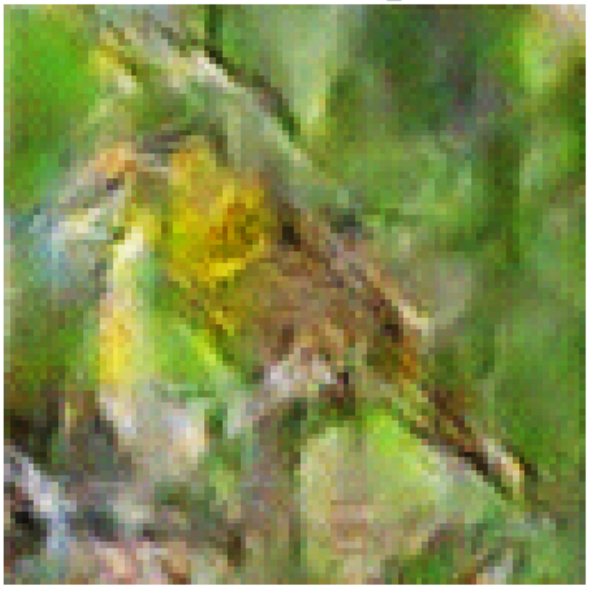
```
C:\Users\aymisxx\AppData\Local\Temp\ipykernel_10568\80036120.py:9: FutureWarnin
g: You are using `torch.load` with `weights_only=False` (the current default va
lue), which uses the default pickle module implicitly. It is possible to constr
uct malicious pickle data which will execute arbitrary code during unpickling
(See https://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models
for more details). In a future release, the default value for `weights_only` wi
ll be flipped to `True`. This limits the functions that could be executed durin
g unpickling. Arbitrary objects will no longer be allowed to be loaded via this
mode unless they are explicitly allowlisted by the user via `torch.serializatio
n.add_safe_globals`. We recommend you start setting `weights_only=True` for any
use case where you don't have full control of the loaded file. Please open an i
ssue on GitHub for any issues related to this experimental feature.
  G.load_state_dict(torch.load(os.path.join(save_dir, "generator.pth"), map_loc
ation=DEVICE))
```

CUB-200: Class-Specific Real vs Generated Comparisons

Real: 182.Yellow_Warbler



Generated: 182.Yellow_Warbler

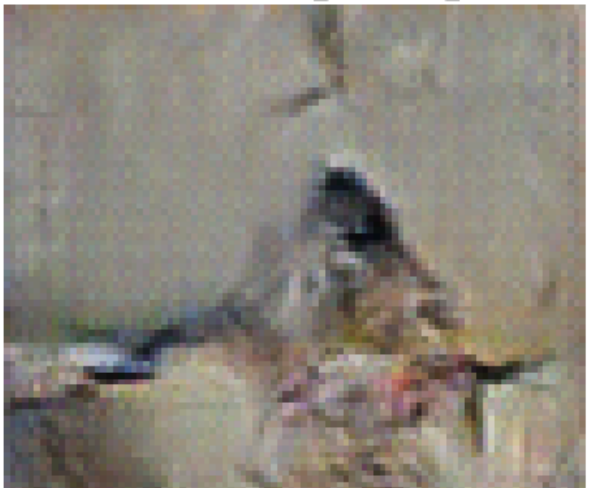


Real: 161.Blue_winged_Warbler

Generated: 161.Blue_winged_Warbler

Real: 114.Black_throated_Sparrow

Generated: 114.Black_throated_Sparrow



Interpretation of CUB-200 Results: Semantic Consistency Under Environmental Complexity

The CUB-200 results demonstrate that the label-conditioned robot vision model is able to capture **class-level semantic cues**, particularly **dominant color distributions and coarse object presence**, despite operating in a highly challenging visual regime.

Unlike MNIST or Oxford-102, the CUB-200 dataset contains:

- complex and highly variable **background elements** (branches, foliage, water bodies, sky, man-made structures),
- significant **pose variation** (perching, flying, standing),
- and fine-grained inter-class differences that are often subtle even for human observers.

Within this context, the generated images consistently reflect **class-relevant color priors** (e.g., yellow hues for Yellow Warbler, blue tones for Florida Jay), indicating that semantic conditioning remains effective. However, background textures and scene statistics are strongly entangled with the foreground object, leading to blurred or abstract representations.

From a robotics perspective, this outcome is expected and informative:

- The model prioritizes **global semantic alignment** over fine anatomical detail.
- Environmental context is treated as part of the visual distribution, mirroring how real robotic sensors often observe objects embedded within cluttered scenes.
- The results highlight the difficulty of fine-grained object synthesis without explicit spatial, part-based, or attention-driven inductive biases.

Overall, these outputs confirm that the vision module learns **meaningful semantic associations** under complex environmental conditions, while clearly exposing the limits of baseline conditional generative models for fine-grained perception tasks.

Final Summary and Takeaways

This project investigated **label-conditioned robot vision** across three progressively challenging perception regimes: MNIST, Oxford-102 Flowers, and

CUB-200 Birds.

Key takeaways:

- **MNIST** validated the core conditioning mechanism, demonstrating reliable semantic controllability in a low-entropy setting.
- **Oxford-102** extended the framework to real-world imagery, showing that the model can learn color- and texture-dominated semantics under moderate visual complexity.
- **CUB-200** served as a stress test for fine-grained perception, revealing that while class-level semantic cues are preserved, detailed structural reconstruction degrades under high semantic cardinality and complex backgrounds.

From a Robotics & Autonomous Systems standpoint, these results are both realistic and valuable. They illustrate how generative perception models behave when confronted with increasing environmental variability and semantic density—conditions typical of real robotic sensing.

The project intentionally emphasizes **interpretability and failure analysis** over photorealism. By exposing where and why semantic conditioning succeeds or breaks down, this work provides a clear foundation for future extensions, including attention-based architectures, diffusion models, and structured spatial priors.

In summary, this study establishes a principled baseline for label-conditioned robot vision and offers concrete insights into the challenges of scaling generative perception to complex, real-world environments.